

Forecasting inflation using disaggregates and machine learning

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Abstract

This paper examines the effectiveness of several forecasting methods for predicting inflation, focusing on aggregating disaggregated forecasts – also known in the literature as the bottom-up approach. Taking the Brazilian case as an application, we consider different disaggregation levels for inflation and employ a range of traditional time series techniques as well as linear and nonlinear machine learning (ML) models to deal with a larger number of predictors. For many forecast horizons, the aggregation of disaggregated forecasts performs just as well survey-based expectations and models that generate forecasts using the aggregate directly. Overall, ML methods outperform traditional time series models in predictive accuracy, with outstanding performance in forecasting disaggregates. Our results reinforce the benefits of using models in a data-rich environment for inflation forecasting, including aggregating disaggregated forecasts from ML techniques, mainly during volatile periods. Starting from the COVID-19 pandemic, the random forest model based on both aggregate and disaggregated inflation achieves remarkable predictive performance at intermediate and longer horizons.

Keywords: inflation forecasting; disaggregated inflation; bottom-up approach; data-rich environment; machine learning.

JEL Codes: C22, C38, C52, C53, C55, E37.

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1 Introduction

Economists and econometricians aim to provide as accurate inflation forecasts as possible by utilizing the most efficient approaches available. An important question is whether considering disaggregated inflation in different markets or economic classifications can enhance the forecasting performance for aggregate inflation. At first, this approach could capture trend dynamics, seasonality, and short-term changes more effectively (Espasa, Senra, and Albacete, 2002). In other words, using subcomponents would allow the econometric models to capture the heterogeneity underlying the aggregate variable better (Bermingham and D’Agostino, 2014). Given an unknown data-generating process, whether direct or indirect forecasting through aggregating disaggregated forecasts can improve or not forecast accuracy is strictly an empirical question (Lütkepohl, 1984; Hendry and Hubrich, 2011; Faust and Wright, 2013). Nevertheless, a critical challenge that emerges is the increase in estimation uncertainty. To mitigate this problem, we implement a disaggregated analysis using machine learning (ML) methods that can deal with the bias-variance trade-off. Studies such as Inoue and Kilian (2008), Garcia, Medeiros, and Vasconcelos (2017), and Medeiros, Vasconcelos, Veiga, and Zilberman (2021) point out the benefits of these techniques for inflation forecasting. Our paper employs these techniques in the context of disaggregated analysis, something scarcely explored in the literature.

The broad literature on inflation forecasting documents that the predictive performance of survey-based forecasts is challenging to beat, especially in the short-term horizons – current and immediate next months (Thomas, 1999; Ang, Bekaert, and Wei, 2007; Croushore, 2010). Faust and Wright (2013) argue that “purely subjective forecasts are in effect the frontier of our ability to forecast inflation” because, besides private sectors and central banks having access to econometric models, they add expert judgment to these models. Consequently, “a useful way of assessing models is by their ability to match survey measures of inflation expectations” (Faust and Wright, 2013). A potential explanation for this phenomenon is that forecasters are likely to have a richer information set than the econometrician employing a standard set of macroeconomic variables as predictors for inflation (Del Negro and Eusepi, 2011). Thus, including revealed expectations among the predictors is a way to exploit an information set that is not available. Baştürk, Çakmaklı, Ceyhan, and Van Dijk (2014), Altug and Çakmaklı (2016), Garcia, Medeiros, and Vasconcelos (2017), Fulton and Hubrich (2021), and Bańbura, Leiva-León, and Menz (2021) find evidence favorable to the incorporation of survey-based forecasts into forecasting econometric models.

This paper examines the effectiveness of various forecasting methods for predicting aggregate inflation, focusing on aggregating disaggregated forecasts – also known in the literature as the bottom-up approach. Using the Brazilian case as an example, we compare the predictive performance of the bottom-up approach with traditional approaches in the literature, including survey-based forecasts and direct forecasting based exclusively on the aggregate. We explore different levels of disaggregation to assess how forecasts based on disaggregate price levels fare relative to those that rely solely on aggregate. Granularity is a potential advantage of considering disaggregates. Besides the specific effects of the traditional macro-variables related to money, economic activity, government, and external sector, we include lagged and crossed effects between disaggregates. When we compute our forecasts, we also consider *available* survey-based expectations as a predictor to add information not captured by other variables. Finally, we employ a range of traditional time series techniques, as well as linear and nonlinear ML techniques to deal with a larger number of predictors. More specifically, we consider these modeling possibilities:

1. Traditional time series methods: random walk (RW), historical mean, and autoregressive (AR) models;
2. Shrinkage-based models with or without sparsity, namely, Ridge and adaptive LASSO (adaLASSO);
3. Factor and target factor augmented models;
4. FarmPredict, a model that bridges both common factor and sparsity structures. With this method, we can explore remaining sparse idiosyncratic effects after controlling for common factors, and autoregressive, expectation and deterministic components;
5. Complete subset regression (CSR), an ensemble method that combines estimates from all possible linear regression models keeping the number of predictors fixed;
6. Random forest (RF), a bagged ensemble of non-linear tree-based models;
7. Model combination via an average of forecasts for each disaggregation.

The Brazilian case is interesting for several reasons. First, the Broad Consumer Price Index (*Índice de Preços ao Consumidor Amplo – IPCA*), which serves as the official Brazilian price index, is available monthly and boasts a rich structure to be explored. The index contains several disaggregation levels and all time-varying weights of goods and services in the representative consumption basket are readily available. Second, the Central Bank of Brazil conducts the Focus survey, an extensive daily survey of expectations for multiple forecast horizons for some variables, including inflation. This survey reflects experts’ opinions, mainly financial market professionals, and may contain private information that is not available to the econometrician. Beyond its utility as a predictor for generating model-based forecasts, Focus’ inflation expectations can be used as a benchmark to assess whether improving survey-based forecasts for a given horizon is possible. Third, due to Brazil’s inflationary history, in addition to the official price index, the country has several price indexes that may be used as predictors for inflation. Hence, it is pertinent to examine whether this information is valuable for forecasting Brazilian inflation over future horizons.

Findings. Among the main results of this paper are:

- (i) It is challenging to outperform the survey-based forecast (Focus) before the COVID-19 pandemic; however, it is achievable post-pandemic (including at short horizons);
- (ii) Taking disaggregated inflation into account tends to generate forecasts as good as survey-based expectations and forecasts based on aggregate inflation directly;
- (iii) Overall, ML methods tend to outperform traditional time series models in predictive accuracy, with outstanding performance in predicting disaggregates;
- (iv) There exists high variability in the type of predictors selected by the adaLASSO and FarmPredict;
- (v) The *available* survey-based inflation expectations and price variables are relevant predictors;
- (vi) Starting from the pandemic, the RF using aggregate inflation or some disaggregation achieves remarkable predictive performance at intermediate and longer horizons.

Contributions for the literature. We can summarize the main contributions of this paper in two fields. First, this paper advances the literature on inflation forecasting via aggregation of disaggregated forecasts by considering many predictors for each disaggregate, as well as several statistical and econometric methods underexplored in this literature. Many papers employ traditional time

series models and a limited number of predictors. In this context, some papers find evidence favoring the bottom-up approach for the Euro Area (Espasa, Senra, and Albacete, 2002; Espasa and Albacete, 2007) and various countries (Bruneau, De Bandt, Flageollet, and Michaux, 2007; Moser, Rumler, and Scharler, 2007; Capistrán, Constandse, and Ramos-Francia, 2010; Aron and Muellbauer, 2012; Carlo and Marçal, 2016; Fulton and Hubrich, 2021). On the other hand, some papers find that aggregating forecasts by components does not necessarily improve aggregate inflation forecasting (Benalal, Diaz del Hoyo, Landau, Roma, and Skudelny, 2004; Hubrich, 2005; Hendry and Hubrich, 2011). In turn, Duarte and Rua (2007), Ibarra (2012), and Bermingham and D’Agostino (2014) highlight the benefits of aggregating a large number of disaggregates. By employing a large number of predictors, Florido (2021, Chapter 1) points out the benefits of the disaggregated analysis in inflation nowcasting, while Araujo and Gaglianone (2023) do not find good results by using a disaggregation in multi-period forecasting. We show that the bottom-up approach can generate multi-period forecasts as accurately as survey-based expectations and direct forecasts.

Second, our analysis extends the literature on machine learning (ML) benefits to forecasting inflation by showing a useful application of these methods considering the aggregation of disaggregated forecasts in a data-rich environment. The employ of ML methods to directly forecast inflation started with factor and principal component models (Stock and Watson, 1999, 2002; Forni, Hallin, Lippi, and Reichlin, 2003; Bai and Ng, 2008; Ibarra, 2012), and neural network models (Moshiri and Cameron, 2000; Nakamura, 2005; Choudhary and Haider, 2012). Several other papers expanded the list of methods to shrinkage-based models (e.g., Ridge and LASSO), Bayesian methods, bagging, boosting, random forest (RF), and complete subset regressions (CSR), but keeping focus on forecast inflation directly from the aggregate (Inoue and Kilian, 2008; Medeiros, Vasconcelos, and Freitas, 2016; Garcia, Medeiros, and Vasconcelos, 2017; Zeng, 2017; Baybuza, 2018; Medeiros, Vasconcelos, Veiga, and Zilberman, 2021; Araujo and Gaglianone, 2023). Florido (2021, Chapter 1) considers ML techniques, disaggregated inflation, and a broad set of predictors in inflation nowcasting, finding good results. Araujo and Gaglianone (2023) consider the inflation disaggregated into administered prices, services, industrial goods, and food at home to generate multi-horizon inflation forecasts employing several ML models. However, their results are not favorable to the bottom-up approach. In contrast, our combination between disaggregated analysis, ML, and many predictors yields promising results and opens up new possibilities for further exploration.

We also point out five other minor contributions. First, we corroborate the findings of Baştürk, Çakmaklı, Ceyhan, and Van Dijk (2014), Altug and Çakmaklı (2016), Garcia, Medeiros, and Vasconcelos (2017), Fulton and Hubrich (2021), and Bańbura, Leiva-León, and Menz (2021) regarding the benefits of incorporating a survey-based expectation as a predictor when econometrician computes their forecasts. The presence of this variable is relevant to improve predictive accuracy even for some disaggregation levels. Second, when estimating a factor-augmented autoregression model using a method that allows predictor selection, we find that the factor that summarizes most of the predictors’ variability is not relevant for predicting inflation. Hence, using an estimation method such as the adaLASSO or the approaches of Bai and Ng (2008, 2009) instead of least squares may be beneficial. Third, our paper is one of the first to employ the FarmPredict, a model proposed by Fan, Masini, and Medeiros (2021) that combines factor and sparse linear regressions. We adapt it to allow the simultaneous estimation via adaLASSO of a final model containing lags, common factors, and idiosyncratic components. Fourth, as in Duarte and Rua (2007), Ibarra (2012), and Bermingham and D’Agostino (2014), we also indicate the potential benefits of considering a high level of disaggre-

gation. However, there are caveats about how to improve the bottom-up approach by considering different models predicting different disaggregates. Finally, our analysis underscores the importance of examining sub-periods and emphasizing the benefits of model-based forecasts in volatile periods, as also pointed out by [Altug and Çakmaklı \(2016\)](#) and [Medeiros, Vasconcelos, Veiga, and Zilberman \(2021\)](#).

Outline. This paper has five more sections in addition to this Introduction. Section 2 presents the forecasting methodology, and Section 3 describes the models, estimation, metrics, and test to compute and assess the results. Section 4 displays the data and setup. Section 5 presents the results and provides an economic discussion about them. Finally, Section 6 concludes. Appendixes from A to D offer supplementary information and complementary results.

2 Forecasting methodology

2.1 “Traditional” inflation forecasting

Let π_t be the (aggregate) inflation at period t . We compute the inflation from the percentage change in a price index based on a typical consumption basket. For forecasting purposes, assume there are J predictors for inflation. Let $\mathbf{z}_t$ be a J -dimensional vector of these explanatory variables *observed* at t , that is, the information set available to the econometrician to perform the forecasting. Notice that $\mathbf{z}_t$ can contain both the last available realizations of the predictor variables as well as lags of these variables. Lastly, let $\mathcal{M}_{t,h}$ be a time-varying mapping between explanatory variables (predictors) and inflation h periods ahead. As the estimation is based on moving windows, the mapping is dependent on time, which we indicate by the subscript t .

There are several possibilities to estimate the mapping $\mathcal{M}_{t,h}$. Initially, we choose between linear or non-linear specifications. In a rich-data environment, we can consider dimensionality reduction or shrinkage with or without selecting predictors. Whatever the choices, we must be careful to avoid overfitting. Finally, an h -period-ahead forecast is given by

$$\hat{\pi}_{t+h|t} = \hat{\mathcal{M}}_{t,h}(\mathbf{z}_t)$$

where hats indicate estimation.

2.2 Aggregation of disaggregated forecasts

Besides the general price index and their percentage change, the aggregate inflation π_t , now consider the availability of N^d disaggregated price indexes (subcomponents of the original price index) indexed by $i = 1, \dots, N^d$. The letter d indicates the disaggregation level. Let π_{it}^d be the percentage change of the disaggregate i in disaggregation level d at period t . Let ω_{it}^d be the weight of the disaggregate i at disaggregation level d in the general price index at period t . Note that these weights are time-varying since the composition of the representative consumption basket may change over time. The relationship between inflation and price changes in disaggregated indexes is given by

$$\pi_t = \sum_{i=1}^{N^d} \omega_{it}^d \pi_{it}^d, \quad (2.1)$$

that is, aggregate inflation is a weighted average of “disaggregated inflations” (price changes).

Let $\mathbf{z}_t^d$ be a J^d -dimensional vector of *all* explanatory variables for price changes *observed* at t by the econometrician at disaggregation level d . Note that it is expected that $J^d > J$ since in the disaggregated case we potentially have more information: in addition to all the other explanatory variables available in the aggregated case, we can use the lagged price changes of the other disaggregates as predictors for a specific disaggregate. The question arises as to whether capturing and exploring the crossed dependence between disaggregated prices could enhance inflation forecasting. Let $\mathcal{M}_{i,t,h}^d$ be a time-varying mapping between predictors and h -period-ahead price variation of each disaggregate $i = 1, \dots, N^d$ at disaggregation level d . Following (2.1), an h -period-ahead forecast for aggregate inflation is given by

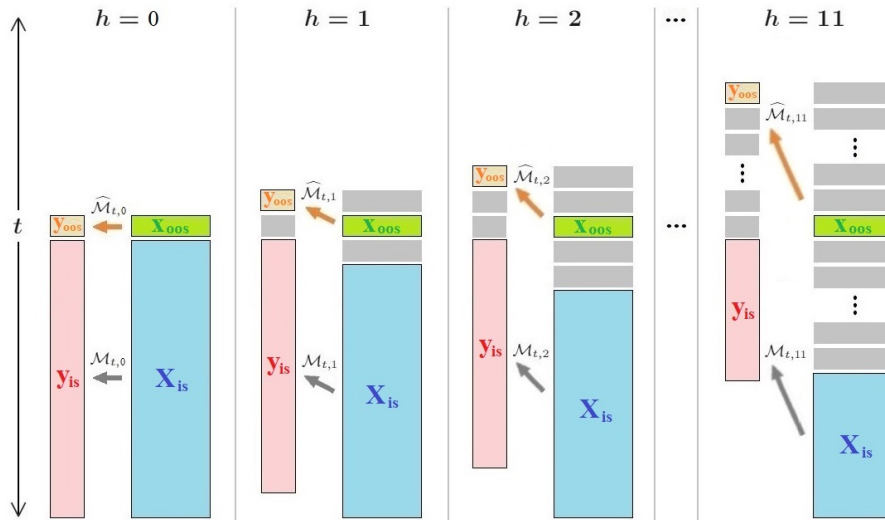
$$\hat{\pi}_{t+h|t} = \sum_{i=1}^{N^d} \tilde{\omega}_{it}^d \hat{\pi}_{i,t+h} = \sum_{i=1}^{N^d} \tilde{\omega}_{it}^d \hat{\mathcal{M}}_{i,t,h}^d(\mathbf{z}_t^d),$$

where $\tilde{\omega}_{it}^d$ is the weight of the disaggregate i at disaggregation level d in aggregate index *observed* at t by the econometrician, that is, the *last available* weight at period t and not the weight *evaluate* for the period t – which we previously indicate simply by ω_{it}^d .

2.3 Direct forecasting approach and expanding window scheme

We employ a direct forecast approach considering expanding windows for (monthly) horizons $h \in \{0, 1, \dots, 11\}$. We take the time-adjusted predictors to fit the mapping between them and inflation in this approach. For example, suppose we want to generate a forecast for the current period ($h = 0$), which is called *nowcasting*. In that case, we consider the most recently *available* information to estimate the desired mapping. Conversely, when computing a one-month-ahead forecast ($h = 1$), we use the information available up until the preceding period in which the forecast is estimated. We continue this way until we calculate the forecast for $h = 11$, utilizing information *available* ten periods prior. Figure 1 illustrates the exercise. Following the computation of forecasts based on a given period, we advance the time window by one period and repeat the estimation procedure for each forecast horizon, subsequently calculating new forecasts.

Figure 1: Direct forecasting approach with expanding window scheme



Notes: y indicates the target variable. X represents predictor variables. Subscripts “is” and “oos” denote in-sample and out-of-sample, respectively.

3 Models and forecast evaluation

3.1 Models

To enhance clarity in presenting the following forecast methods, we omit the superscript d that indicates the level of disaggregation, whenever applicable.

3.1.1 Benchmarks

Random walk (RW). Considering the aggregated case, the forecast of the h -period-ahead inflation at period t is given by current inflation, that is, $\hat{\pi}_{t+h|t}^{\text{RW}} = \pi_t$.

Historical mean. Also for the aggregated case, a prediction for h periods ahead is given by historical average inflation computed at t , that is,

$$\hat{\pi}_{t+h|t}^{\text{Hist. Mean}} = \bar{\pi}_{t+h|t} = \frac{1}{S} \sum_{s=t-S+1}^t \pi_s$$

where S is the number of previously observed inflation measures (expanding window length).

Autoregressive model – AR(p). For both aggregated and disaggregated cases, in the direct forecast approach, for each horizon h , we can be written a p -order AR model as

$$\pi_t = \mu + \sum_{l=1}^p \phi_l \pi_{t-h-l+1} + \varepsilon_t$$

where ε_t is an error term. The order p can be previously fixed or selected via some information criterion (e.g., BIC). Thus, a h -period-ahead inflation forecast is given by

$$\hat{\pi}_{t+h|t}^{\text{AR}} = \hat{\mu} + \sum_{l=1}^p \hat{\phi}_l \pi_{t-h-l+1}.$$

where $\hat{\mu}$ and $\hat{\phi}$'s are least squares (OLS) estimates.

Augmented autoregressive model. Including seasonal dummies and inflation expectation, we can write the model

$$\pi_t = \mu + \sum_{l=1}^p \phi_l \pi_{t-h-l+1} + \eta \pi_{t|t-h}^e + \sum_{m=1}^{11} \delta_m d_{mt} + \varepsilon_t \quad (3.1)$$

where $\pi_{t|t-h}^e$ is the inflation expectation for the period t available at $t-h$, d_{mt} is a seasonal dummy that assumes value 1 for month m , and δ_m is a coefficient associated with seasonal dummy d_{mt} . In this framework, we estimate the coefficients via OLS, and a h -period-ahead forecast is given by

$$\hat{\pi}_{t+h|t}^{\text{Aug. AR}} = \hat{\mu} + \sum_{l=1}^p \hat{\phi}_l \pi_{t-l+1} + \hat{\eta} \pi_{t+h|t}^e + \sum_{m=1}^{11} \hat{\delta}_m d_{m,t+h}.$$

(Empirical) Hybrid New Keynesian Phillips curve (HNKPC). Following and *adapting* price-setting models such as those presented in [Galí and Gertler \(1999\)](#) and [Blanchard and Galí \(2007\)](#), we employ

a forecasting model for the aggregate inflation based on a hybrid Phillips curve given by

$$\pi_t = \mu + \sum_{l=1}^p \phi_l \pi_{t-h-l+1} + \eta \pi_{t|t-h}^e + \psi_1 g_{t-h} + \psi_2 \Delta s_{t-h} + \varepsilon_t$$

where g_{t-h} is some economic activity measure *observed* at $t-h$, and Δs_{t-h} is an exchange rate measure *observed* at $t-h$. We compute the forecast by

$$\hat{\pi}_{t+h|t}^{\text{HNKPC}} = \hat{\mu} + \sum_{l=1}^p \hat{\phi}_l \pi_{t-l+1} + \hat{\eta} \pi_{t+h|t}^e + \hat{\psi}_1 g_t + \hat{\psi}_2 \Delta s_t$$

where $(\hat{\mu}, \hat{\phi}, \hat{\eta}, \hat{\psi}_1, \hat{\psi}_2)$ are OLS estimates.

3.1.2 Shrinkage-based models

Ridge (with incomplete information). For disaggregated cases, we consider the augmented AR model (3.1) with the addition of other lagged disaggregates:

$$\pi_{it} = \mu + \sum_{i'=1}^{N^d} \sum_{l=1}^p \phi_{i'l} \pi_{i',t-h-l+1} + \eta_i \pi_{t|t-h}^e + \sum_{m=1}^{11} \delta_{im} d_{mt} + \varepsilon_{it} \quad i = 1, \dots, N^d,$$

where N^d is the number of subcomponents in the disaggregation level indicated by d . We consider four disaggregation levels in this paper: aggregate inflation, economic categories defined by the BCB, and groups and subgroups from IPCA (IBGE).

We estimate the coefficients employing the Ridge estimator:

$$(\hat{\mu}_i, \hat{\beta}_{\text{Ridge},i}(\lambda)) = \underset{\mu_i, \beta_i}{\operatorname{argmin}} \left\{ \frac{1}{T-h} \sum_{t=1}^{T-h} (\pi_{it} - \mu_i - \beta_i \mathbf{z}_{t-h})^2 + \lambda_i \sum_{j=1}^J \beta_{ij}^2 \right\}$$

where λ_i is a regularization parameter, $\mathbf{z}_{t-h}$ is a vector with all predictors, and β_i is a vector of coefficients. Chosen λ via information criteria (e.g., BIC), a prediction for h periods ahead is given by

$$\hat{\pi}_{t+h|t}^{\text{Ridge}} = \sum_{i=1}^{N^d} \omega_{it} \hat{\pi}_{i,t+h|t}^{\text{Ridge}} \quad \text{with} \quad \hat{\pi}_{i,t+h|t}^{\text{Ridge}} = \hat{\mu}_i + \hat{\beta}_{\text{Ridge},i} \mathbf{z}_t.$$

adaLASSO (with full information). For all cases, consider the model with full information given by

$$\pi_{it} = \mu_i + \sum_{i'=1}^{N^d} \sum_{l=1}^p \phi_{i'l} \pi_{i',t-h-l+1} + \eta_i \pi_{t|t-h}^e + \sum_{m=1}^{11} \delta_{im} d_{mt} + \sum_{j=1}^J \sum_{l=1}^p \theta_{ijl} x_{j,t-h-l+1} + \varepsilon_{it}$$

where $\mathbf{x}_{t-h} \in \mathbb{R}^{J \cdot p}$ is an expanded vector of potential predictors for π_{it} . We estimate this model employing the adaptive LASSO (adaLASSO). Introduced by [Zou \(2006\)](#), this method selects predictors and their optimization problem is given by

$$(\hat{\mu}_i, \hat{\beta}_{\text{adaLASSO},i}(\lambda, \omega)) = \underset{\mu_i, \beta_i}{\operatorname{argmin}} \left\{ \frac{1}{T-h} \sum_{t=1}^{T-h} (\pi_{it} - \mu_i - \beta_i \mathbf{z}_{t-h})^2 + \zeta \sum_{j=1}^J \zeta_{ij} |\beta_{ij}| \right\}$$

where ζ is a regularization parameter, $\mathbf{z}_{t-h} \in \mathbb{R}^V$, $V = N^d \cdot p + 12 + J \cdot p$, is a vector of *all* predictors, and $\boldsymbol{\zeta} = (\zeta_1, \dots, \zeta_V)$ is a vector of weights obtained previously employing LASSO – a estimator that assumes $\zeta_{ij} = 1$, for all j . More precisely, we compute the weights via

$$\zeta_{ij} = \left(\left| \hat{\beta}_{\text{LASSO},ij} \right| + \frac{1}{\sqrt{T}} \right)^{-1},$$

where we add $T^{-1/2}$ to allow a variable that is not selected in the first stage to have a chance of being selected in the second stage.

As before, a h -period-ahead forecast is $\hat{\pi}_{i,t+h|t}^{\text{adaLASSO}} = \hat{\mu}_i + \hat{\beta}_{\text{adaLASSO},i} \mathbf{z}_t$.

3.1.3 Factor models

(Augmented) factor model. Consider that all regressors are normalized for both aggregate and disaggregate cases. Thus, for $i = 1, \dots, N^d$, a factor-augmented autoregression model is described by

$$\begin{aligned} \mathbf{x}_t &= \sum_{k=1}^K \lambda_k f_{kt} + \mathbf{u}_t \\ \pi_{it} &= \mu_i + \sum_{i'=1}^D \sum_{l=1}^p \phi_{i'l} \pi_{i',t-l+1} + \eta_i \pi_{t|t-h}^e + \sum_{m=1}^{11} \delta_{im} d_{mt} + \sum_{k=1}^K \beta_{ik} \hat{f}_{k,t-h} + \varepsilon_{it} \end{aligned} \quad (3.2)$$

from which we compute common factors $\hat{\mathbf{f}}_t = (\hat{f}_{1t}, \dots, \hat{f}_{Kt})$ and factor loadings $\lambda_k = (\lambda_{1k}, \dots, \lambda_{Jk})$ by combining principal component analysis (PCA) and OLS. Finally, we compute $(\hat{\mu}_i, \hat{\boldsymbol{\phi}}, \hat{\eta}_i, \hat{\boldsymbol{\beta}})$ via adaLASSO.

For identification purposes, we assume that

$$\mathbb{E}(\mathbf{f}_t | \mathbf{u}_t) = 0, \quad \text{Cov}(\mathbf{u}_t, \varepsilon_t) = 0, \quad \text{Var}(\mathbf{f}_t) = \mathbf{I}_K, \quad \text{Var}(\mathbf{u}_t) = \boldsymbol{\Omega} = \text{diag}(\sigma_1^2, \dots, \sigma_p^2).$$

The number of factors K is selected via information criterion IC_{p2} of [Bai and Ng \(2002\)](#), and the forecast h periods ahead is given by

$$\hat{\pi}_{i,t+h|t}^{\text{Factor}} = \hat{\mu}_i + \sum_{i'=1}^D \sum_{l=1}^p \hat{\phi}_{i'l} \pi_{i',t-l+1} + \hat{\eta}_i \pi_{t+h|t}^e + \sum_{m=1}^{11} \hat{\delta}_{im} d_{m,t+h} + \sum_{k=1}^K \hat{\beta}_{ik} \hat{f}_{kt}$$

where $\hat{f}_{kt}$ is the k -th factor evaluated at t .

Target factor model. Proposed by [Bai and Ng \(2008\)](#), in this “hard thresholding” version, this approach controls for the participation of normalized explanatory variables in the factor construction. In a previous stage, for each predictor indexed by $j = 1, \dots, J$, and disaggregate indexed by $i = 1, \dots, N^d$, we estimate

$$\pi_{it} = \mu_i + \sum_{i'=1}^{N^d} \sum_{l=1}^p \phi_{i'l} \pi_{i',t-l} + \eta_i \pi_{t|t-h}^e + \sum_{m=1}^{11} \delta_{im} d_{mt} + \theta_{ij} x_{j,t-h} + \nu_{it}$$

and run the hypothesis test $\theta_{ij} = 0 \times \theta_{ij} \neq 0$ for some significance level α . If θ_{ij} is statistically different from zero, we employ x_j in the factor estimation. Let $\mathbf{x}_t(\alpha, i)$ be the set of selected variables for i -th disaggregation. Finally, we proceed as in the traditional factor-augmented autoregressive

model: we perform

$$\mathbf{x}_t(\alpha, i) = \sum_{k=1}^K \lambda_k f_{kt} + \mathbf{u}_t$$

which $\hat{f}_t$ and $\hat{\lambda}_k$ are computed via PCA and OLS. Then we estimate the augmented (target) factor model via adaLASSO and compute the forecast as before.

3.1.4 FarmPredict

Some idiosyncratic errors of the factor model, that is, some $\mathbf{u}_t$ entries in Equation (3.2), can impact the price variation, which the common factor structure does not capture. Defining $\hat{\mathbf{u}}_t = \mathbf{x}_t - \sum_{k=1}^K \hat{\lambda}_k \hat{f}_{k,t}$, a J -dimensional vector, we can introduce lags of $\mathbf{u}_t$ on the factor model:

$$\begin{aligned} \pi_{it} = & \mu_i + \sum_{i'=1}^{N^d} \sum_{l=1}^p \phi_{i'l} \pi_{i',t-h-l+1} + \gamma_i \pi_{t|t-h}^e + \sum_{m=1}^{11} \delta_{im} d_{mt} \\ & + \sum_{k=1}^K \sum_{l=1}^p \beta_{ikl} \hat{f}_{k,t-h-l+1} + \sum_{j=1}^J \sum_{l=1}^p \theta_{ijl} \hat{u}_{j,t-h-l+1} + \varepsilon_{it}. \end{aligned} \quad (3.3)$$

This model is a specific form of a general model called FarmPredict proposed by [Fan, Masini, and Medeiros \(2021\)](#). Here, we estimate the “final equation” (3.3) with all regressors *simultaneously* employing the adaLASSO. Next, we compute the forecast.

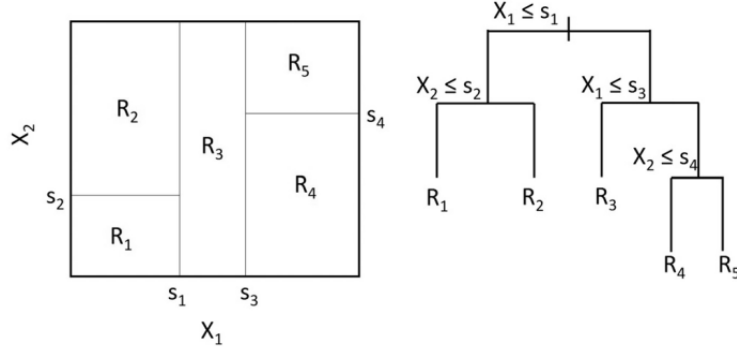
3.1.5 Complete subset regression (CSR)

Introduced by [Elliott, Gargano, and Timmermann \(2013, 2015\)](#), this ensemble method combines estimates from all (or several) possible linear regression models, keeping the number of predictors fixed. Let p be the total available predictors and $k \leq p$ be the number of “selected” predictors (complete subsets). The CSR involves the estimation of $\frac{k!}{(k-p)!k!}$ linear models. Variables when “non-selected” has their coefficients set to zero. The final CSR estimate is the average of all estimates. Thus, subset regression has a shrinkage interpretation since when averaging parameters that sometimes assume zero value, this average generates shrunken estimates of the coefficients, which can contribute to more accurate forecasts. Due to the high computational cost arising from a large number of predictors, we (pre-)select $\tilde{p} \leq p$ predictors based on a ranking of t-statistics in absolute value, as in [Garcia, Medeiros, and Vasconcelos \(2017\)](#) and [Medeiros, Vasconcelos, Veiga, and Zilberman \(2021\)](#). This procedure is similar to that used in the target factor model. So, instead of considering all available p predictors, we run the CSR considering $\tilde{p}$ pre-selected predictors.

3.1.6 Random forest (RF)

[Breiman \(2001\)](#) introduces the random forest (RF), a model that combines several based-tree regressions using bagging. A regression tree is a nonparametric model that approximates an unknown nonlinear function with local predictions via recursive partitioning, as illustrated in Figure 2.

Figure 2: A regression tree with two explanatory variables (X_1, X_2)



Notes: s_i , $i = 1, \dots, 4$, indicate splits, while R_k , $k = 1, \dots, 5$, denote regions. Example extracted from [Medeiros, Vasconcelos, Veiga, and Zilberman \(2021\)](#).

Formally, a regression tree model can be written as follows:

$$\pi_{it} = \sum_{k=1}^K c_k \mathcal{I}_k(\mathbf{x}_{t-h} \in R_k)$$

where $\mathcal{I}_k(\mathbf{x}_{t-h} \in R_k)$ is an indicator function that assumes the value 1 when $\mathbf{x}_{t-h}$ belongs to the k -th region R_k , and c_k is the average of π_t in this region. We have to set the minimum number of observations per region. Then, we obtain B trees by implementing a double draw: we draw on the observation dimension using block bootstrap, and we draw variables to incorporate in the estimation of the tree. The idea is that this double draw will ensure the variability of the trees. Let K_b be the number of regions of the b -th tree, $b = 1, \dots, B$. Lastly, the final forecast is given by the average of the forecasts obtained by each tree evaluated in the original data, that is,

$$\hat{\pi}_{i,t+h|t}^{\text{RF}} = \frac{1}{B} \sum_{b=1}^B \sum_{k=1}^{K_b} \hat{c}_{k,b} \mathcal{I}_{k,b}(\mathbf{x}_{t-h} \in R_{k,b})$$

where $R_{k,b}$ is the k -th region of the b -th tree.

3.2 Model combinations via average of forecasts

Methods may perform differently for distinct disaggregates or even over time for the same disaggregate. To mitigate instabilities associated with some method for some disaggregate or at some point in time, for each disaggregation, we will compute a combined forecast given by the average of forecasts generated by all methods applied to this disaggregation and Focus expectations *available* when the econometrician computes their forecasts, that is,

$$\hat{\pi}_{t+h|t}^{\text{Comb},d} = \frac{1}{M^d + 1} \left(\sum_{m=1}^{M^d} \sum_{i=1}^{N^d} \tilde{\omega}_{it}^d \hat{\pi}_{i,t+h|t}^{m,d} + \pi_{t+h|t}^e \right),$$

where d indicates one of four possible disaggregations levels addressed in this paper (aggregate inflation, BCB categories, IBGE groups, and IBGE subgroups), m indicates a method, M^d is the number of methods employed to forecast the inflation for the disaggregation d , N^d is the number of disaggregates in the disaggregation d , and $\tilde{\omega}_{it}^d$ is the weight of disaggregate i of the disaggregation level d in the aggregate index *observed* at t by the econometrician. The idea is to investigate whether this simple combination leads to improvements in forecast performance.

3.3 Evaluation: metrics and test

Metrics. We use out-of-sample root mean squared error (RMSE) as the main metric to evaluate the forecast performance. For each horizon h , this metric is described by

$$\text{RMSE}_h^{m,d} = \left[\frac{1}{T} \sum_{t=1}^T (\pi_{t+h} - \hat{\pi}_{t+h|t}^{m,d})^2 \right]^{1/2}$$

where $\hat{\pi}_{t+h|t}^{m,d}$ indicates a forecast generated by the model m considering the disaggregation level d . The smaller the $\text{RMSE}_{h,m}$, the better the model's predictive performance. For the Diebold-Mariano test, we consider the mean squared error (MSE) defined by

$$\text{MSE}_h^{m,d} = \frac{1}{T} \sum_{t=1}^T (\pi_{t+h} - \hat{\pi}_{t+h|t}^{m,d})^2.$$

Test. To assess the results, we consider the widely employed test developed by [Diebold and Mariano \(1995\)](#). Let $\hat{v}_{t+h|t}^m = \pi_{t+h} - \hat{\pi}_{t+h|t}^m$ be a forecast error of the model m . Here, we omit the disaggregation level d . Let $g(\cdot)$ be a metric to be applied to $\hat{v}_{t+h|t,m}$ (e.g., MSE). The Diebold-Mariano (DM) test statistic is given by

$$d_{m,m'} = \frac{1}{T} \sum_{t=1}^T \left(g(\hat{v}_{t+h|t}^m) - g(\hat{v}_{t+h|t}^{m'}) \right)$$

where m' indicates another model, a competitor (i.e., a benchmark model or specific forecast, for example). We will consider that the normality of DM statistics is likely a trustworthy approximation, including for model-based forecasts.

4 Data and setup

Data. We analyze the period from January 2004 to June 2022, totalizing 18,5 years of monthly data. For aggregate inflation, we employ the IPCA, the official Brazilian price index computed by the Brazilian Institute of Geography and Statistics (*Instituto Brasileiro de Geografia e Estatística* – IBGE). For disaggregations, we consider all groups and subgroups of the IPCA. There are nine groups and 19 subgroups throughout the period analyzed. Subgroups are subdivisions and, in some cases, the group itself. For definition of groups and subgroups, and their respective average weights in the IPCA, see Table A1 in Appendix A. In addition, we use a disaggregation defined by the Central Bank of Brazil (BCB) based on IBGE data. The BCB disaggregation consists of administrated, non-tradables, and tradables items. The use of this last disaggregation is interesting because, in principle, it presents more economic intuition, which can contribute to better forecast performance.

We consider inflation expectations of the Central Bank of Brazil's Focus survey and lags of the predicted variables among the admissible predictors. To forecast a disaggregate, we consider lags of other disaggregates in the same disaggregation, which allows capturing potential lagged "cross-effects". The Focus survey has a daily frequency and contains inflation expectations formed by many economic agents (experts) for several horizons (months) ahead. Reflecting the opinion of experts, the Focus may contain private information that is not available to the econometrician – hence the impor-

tance of considering this variable in our information set. We consider the latest available inflation expectation for the horizon of interest when we generate our forecast. Moreover, there are eighty-nine other predictor variables (and their lags) divided into ten categories: prices and money (17), commodities prices (4), economic activity (19), employment (5), electricity (4), confidence (3), finance (12), credit (4), government (12), and exchange and international transactions (9). In Appendix B, Table B1 presents a description of these variables, the delay for each to become *available* and transformations implemented to guarantee the stationarity. We structure our dataset to closely approximate real-time data for the Brazilian case. The main challenge lies in our lack of access to the first releases of some economic activity variables, including the IBC-Br (a Brazilian Economic Activity Index) and industrial production (and their subcomponents).

Setup. The reference day to compute our forecasts is the last business day of each month. For the results shown in the following section, we consider three lags for all predictive variables, including variables mentioned above, factors in factor models, idiosyncratic components in FarmPredict, and lags of aggregate and all disaggregates. The only exception is the factors in the target factor model for which we employ only one (target) factor. As mentioned in Subsection 2.3, the main results are generated based on expanding windows. In this setup, we generate 114 forecasts for each horizon. The regularization parameters (λ 's) of the Ridge, LASSO, and adaLASSO are obtained via Bayesian Information Criterion (BIC). We restrict the number of possible selected variables by the ceiling of $\sqrt{T}$ to enforce discipline. The number K of latent factors in factor models is selected via Bai and Ng (2002) information criterion IC_{p2} . For CSR, we set $\tilde{p} = 20$ (number of pre-selected predictors) and $p = 4$ (number of selected variables by CSR). For pre-selecting of both target factor and CSR models, we adopt the 5% significance level ($\alpha = 0.05$). In its turn, for the RF models, we allow the trees to grow until five observations by leaf. We set the proportion of selected variables in each split to 1/3 and the number of bootstrap samples to 500 ($B = 500$). All settings are similar to those adopted by Garcia, Medeiros, and Vasconcelos (2017) and Medeiros, Vasconcelos, Veiga, and Zilberman (2021). Finally, to estimate the empirical Phillips curve, we use the Central Bank of Brazil's economic activity index (IBC-Br) and BIS' real effective exchange rate (REER) as a proxy for economic activity and exchange rate, respectively.

5 Results

5.1 Entire period: forecasts from January 2014 to June 2022

Table 1 exhibits the results of forecast performance in terms of root mean squared errors (RMSE) for different models and horizons ranging from nowcasting ($h = 0$) to eleven months ahead ($h = 11$), as well as for 12-month accumulated inflation. We normalize every RMSE to relative terms by computing their ratio to the RMSE of the Focus consensus – the median expectation of the available Focus survey. Thus, a value lower than one indicates that a model numerically outperforms the Focus consensus, while a value greater than one suggests underperformance compared to the same benchmark. At this first moment, the results consider the entire period for which we compute predictions, from January 2014 to June 2022. Each panel of Table 1 considers a group of competitors. In panel A, we have the available and *ex-post* Focus, the latter released by the Central Bank in the following week reflecting the experts' opinions on the same day we compute our forecasts. We note virtually no difference between the available and *ex-post* Focus for longer horizons. However, in the short

term ($h \leq 3$), there is evidence that ex-post Focus statistically outperforms available Focus. Despite being only a few days apart, the informational gain is considerable for shorter horizons, which does not occur for more distant periods since it is unlikely that very relevant information about them will emerge within a few days.

Table 1: Out-of-sample RMSE with respect to the available Focus: Jan/2014 to Jun/2022

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	$\Sigma 12m$
A. Survey													
Focus (available)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Focus (<i>ex-post</i>)	0.933 ***	0.972***	0.993***	1.001	1.000	1.000	1.000	0.999	0.999	1.000	1.001	1.001	0.999
B. Aggregate inflation													
RW	2.782	1.471	1.212	1.251	1.279	1.271	1.234	1.180	1.087	1.063	1.102	1.138	1.485
Historical Mean	3.117	1.342	1.084	1.053	1.034	1.018	0.998	0.990	0.987	0.983	0.982	0.984	1.041
AR	2.534	1.260	1.053	1.077	1.090	1.064	1.008	0.961	0.939*	0.941*	0.964	0.988	0.987
HNKPC	0.946 **	0.957 **	0.972*	0.981	1.036	1.030	0.980	0.968	0.963*	0.974	0.998	1.021	0.984
Augmented AR	0.970	0.981	0.989	1.012	1.023	0.999	0.963	0.943 *	0.936 **	0.961	0.993	1.016	0.954*
adaLASSO	0.944 **	0.948 ***	0.978*	0.993	0.977 **	0.989	0.976	0.976	0.989	1.004	1.043	0.992	0.961*
Factor	0.943 **	0.931 ***	0.966 **	0.964 **	0.974	0.980	0.989	0.986	1.016	1.016	1.052	1.045	0.989
FarmPredict	0.944 **	0.942 ***	0.961 **	0.987	0.999	1.001	1.004	1.020	1.020	1.032	1.027	1.051	1.001
Target Factor	1.280	1.129	1.045	1.155	1.122	1.069	0.937 *	0.989	0.990	0.980	1.019	1.102	0.970
CSR	0.961	0.927 **	0.962 *	0.983	1.022	0.996	0.969	0.940 **	0.940 **	0.975	1.042	1.071	0.935**
Random Forest	1.665	1.110	1.025	1.012	0.996	0.987	0.959	0.949	0.920 **	0.918 **	0.908 **	0.909 **	0.918 *
C. Disaggregation: tradable, nontradable and monitored prices (BCB)													
AR	2.579	1.241	1.039	1.057	1.089	1.085	1.015	1.037	0.990	0.989	1.026	1.036	1.050
Augmented AR	0.978	0.978	1.012	1.061	1.094	1.072	1.010	1.010	0.982	1.003	1.052	1.073	1.048
Ridge	1.018	1.015	1.007	1.019	1.045	1.083	0.999	0.978	0.969	0.960*	1.013	0.981	1.032
adaLASSO	1.099	0.998	0.968	0.969	0.977	0.948 **	0.962*	0.957*	0.979	0.964	0.972	0.957 *	0.903 **
Factor	1.023	0.988	0.991	0.990	0.977	0.934 **	0.940 **	0.977	1.000	1.007	1.009	1.009	0.943*
FarmPredict	1.129	1.024	1.003	0.981	0.969	0.944 **	0.956*	0.984	1.003	1.009	1.008	0.991	0.963
Target Factor	1.331	1.052	0.976	1.090	1.131	1.079	0.943	0.962	0.964	0.991	1.016	1.077	0.936
CSR	2.176	1.178	1.016	1.015	1.018	0.983	0.957	0.977	0.959	0.967	1.022	1.052	0.952
Random Forest	1.934	1.177	1.020	1.051	1.014	1.001	0.976	0.963	0.930 *	0.912 **	0.902 **	0.915 **	0.923
D. Disaggregation: groups (IBGE)													
AR	2.731	1.296	1.048	1.063	1.054	1.032	1.067	1.087	1.118	1.145	1.105	1.095	1.067
Augmented AR	1.004	1.046	1.042	1.056	1.088	1.046	1.067	1.065	1.069	1.105	1.134	1.180	1.028
Ridge	1.656	1.288	1.044	1.032	1.016	1.010	0.986	0.989	0.972	0.983	0.989	0.999	1.005
adaLASSO	1.223	1.090	1.026	1.013	1.003	0.994	1.019	0.989	0.972	1.004	0.988	1.000	0.954
Factor	1.219	1.075	1.034	1.031	1.035	1.021	1.024	0.967	0.974	1.021	1.000	0.990	1.005
FarmPredict	1.334	1.089	1.025	1.049	1.040	1.016	1.023	0.991	0.971	1.029	0.997	0.998	1.020
Target Factor	1.268	1.082	1.068	1.142	1.047	0.997	0.976	0.961	1.003	0.970	1.026	1.092	0.977
CSR	2.146	1.144	1.048	1.047	1.002	0.980	0.940 *	0.926 *	0.975	0.981	0.981	0.985	0.900 *
Random Forest	2.096	1.217	1.026	1.047	1.009	0.998	0.971	0.953	0.944	0.931 *	0.920 **	0.921 *	0.955
E. Disaggregation: subgroups (IBGE)													
AR	3.111	1.502	1.186	1.212	1.244	1.184	1.212	1.253	1.264	1.316	1.275	1.362	1.318
Augmented AR	1.197	1.260	1.290	1.288	1.290	1.191	1.164	1.212	1.239	1.310	1.314	1.440	1.208
Ridge	3.100	1.308	1.058	1.045	1.023	1.026	1.008	1.002	1.000	0.993	0.995	1.016	1.068
adaLASSO	1.380	1.122	1.096	1.056	1.030	1.010	0.990	0.990	0.983	1.011	1.049	1.039	1.005
Factor	1.364	1.147	1.068	1.050	1.039	1.029	1.009	0.985	0.996	1.027	1.040	1.005	1.034
FarmPredict	1.384	1.149	1.047	1.040	1.028	1.008	1.010	0.980	0.982	1.033	1.060	1.042	1.036
Target Factor	1.230	1.059	1.145	1.126	1.117	1.059	0.976	0.937	1.028	0.979	1.051	1.116	1.021
CSR	2.222	1.210	1.076	1.067	1.027	1.028	0.977	0.955	0.939	0.987	1.034	1.030	0.974
Random Forest	2.106	1.232	1.032	1.063	1.021	1.022	0.990	0.962	0.949	0.925 *	0.917 **	0.921 *	0.974
F. Model combinations for disaggregates													
Aggreg. Comb.	1.201	0.989	0.952 **	0.972 *	0.982	0.969	0.942 **	0.934 **	0.935 **	0.939 **	0.971	0.986	0.935 **
BCB Comb.	1.243	1.014	0.969	0.993	0.997	0.977	0.945*	0.954*	0.950*	0.952*	0.975	0.986	0.950
Groups Comb.	1.401	1.068	0.976	1.001	0.988	0.969	0.970	0.954	0.959	0.982	0.978	0.995	0.961
Subgroups Comb.	1.583	1.129	1.035	1.045	1.031	1.007	0.981	0.966	0.985	1.004	1.026	1.046	1.029

Notes: ***, **, and * indicate that for a specific forecast horizon, a model m performed statistically better than the median of the available Focus at 1, 5, and 10% significance levels in a one-tailed Diebold-Mariano test with $H_0 : \text{MSE}(\hat{\pi}_{t+h|t}^m) = \text{MSE}(\pi_{t+h|t}^{\text{Focus}})$ versus $H_1 : \text{MSE}(\hat{\pi}_{t+h|t}^m) < \text{MSE}(\pi_{t+h|t}^{\text{Focus}})$. The value highlighted in bold blue indicates the best model for each horizon in terms of RMSE ratio with respect to *ex-post* Focus, and the values in blue italics indicate the second and third best models.

Panels B to E of Table 1 show the results for each model considering different levels of disaggregation: aggregate inflation, disaggregations from the Central Bank of Brazil (BCB), and disaggregations into IBGE groups and subgroups, respectively. Perhaps not surprisingly, it is hard to outperform the

Focus survey in nowcasting. Exceptions are due to models that forecast aggregate inflation directly. However, such models perform better only than the available Focus. Considering the whole period, no alternative beats the ex-post Focus, which delivers almost 7% RMSE reduction compared to the available survey. However, it is appreciable that some models are competitive with the ex-post Focus. Specialists who report their expectations to the BCB often have access to information unavailable to econometricians, such as private data. Since models do not have this additional information and other advantages, such as including personal judgments, as pointed out by [Faust and Wright \(2013\)](#), their ability to outperform available expectations and get closer to *ex-post* survey-based expectations is a great result. We note that models forecasting disaggregates do not deliver good performance for nowcasting. Lastly, the combinations of models in each level of disaggregation, whose results are shown in Panel F, also do not generate forecasts better.

For other horizons, the contribution of the models becomes more effective. Despite the challenge of surpassing survey-based expectations for short-term horizons such as $h = 1$ and $h = 2$, several models for aggregate inflation (Panel B) achieve good results for these horizons. Like occurred for $h = 0$, the hybrid Phillips curve, adaLASSO, factor model, FarmPredict, and, additionally, the complete subset regression (CSR), deliver the best forecast performances for one and two months ahead. All are statistically superior to the available Focus according to the Diebold-Mariano (DM) test considering at least the more slack significance level (i.e., 10%). Furthermore, these models also numerically outperform the ex-post Focus. On the other hand, the adaLASSO using BCB disaggregation (Panel C) is the only model employing any disaggregation among the best models. However, this model is not statistically superior to the available Focus by the DM test. Finally, regarding these shorter horizons, it is worth highlighting the performance of the average forecast of the models for aggregate inflation, which achieves the highest accuracy for $h = 2$ by presenting a statistically significant reduction of almost 5% in RMSE.

Models considering some disaggregation for the inflation yield better results starting from the 4-month horizon. The adaLASSO, factor model, and FarmPredict, all using the BCB disaggregation, perform well for forecast horizons ranging from fourth to seventh months. These models are statistically superior to available or ex-post Focus at various periods. Regarding the use of disaggregated inflation data in groups from the IBGE, it is worth mentioning the good performance of the CSR, which achieves the best result among all the options for $h = 7$. Another highlight is the combination of forecasts generated by models that directly forecast the aggregate inflation, which achieves a statistically significant reduction of 6% in RMSE from 6 to 9 months ahead. For $h \geq 8$, there is broad dominance of the random forest (RF), whether using aggregate inflation or some disaggregation. Frequently, for these more distant horizons, the RF registers a statistically significant reduction in RMSE ranging from 7% to 10% in comparison to the survey-based expectations. This result highlights that the RF, employing IBGE group disaggregation, achieves the best performance among all competitors for inflation accumulated over 12 months (see last column of Table 1), closely followed by the adaLASSO using BCB disaggregation, which achieves a similar RMSE reduction.

Remarks. Considering the forecast performance of various models from January 2014 to June 2022, we observe that different approaches are more effective at different times. In the short term, machine learning models that deal directly with aggregate inflation perform better, whereas for intermediate horizons of 4 to 7 months, considering the BCB disaggregation lead to significant benefits. For the period between 6 and 9 months ahead, the average of forecasts obtained from models that used

only aggregate inflation also perform well. Finally, for longer horizons of 8 months or more, regardless of the approach, the RF delivers the best forecast performances. While [Garcia, Medeiros, and Vasconcelos \(2017\)](#) points to the superiority of the CSR in several horizons, we only verify the prevalence of the CSR for $h = 1$ and other isolated good performances. The RF's performance in predicting inflation had already been pointed out by [Medeiros, Vasconcelos, Veiga, and Zilberman \(2021\)](#) when analyzing the case of the United States and highlighting the benefits of this method for dealing with non-linearities. The advent of the COVID-19 pandemic in Brazil changes the price dynamics considerably from 2020 onwards. Because of that, in what follows, we divide our analysis into two sub-periods: (i) before the pandemic, from January 2014 to February 2020, and (ii) after the pandemic, from March 2020 onwards.

5.2 Forecasts before and after of COVID-19 pandemic

Table 2 shows that the sub-period between January 2014 and February 2020 is quite challenging for model-based forecasts. Almost no RMSE ratios are below 1, with exceptions mainly in the short term. Nevertheless, no model is able to beat the ex-post Focus or be statistically superior to the available Focus for nowcasting. For $h = 1$, only the factor model and FarmPredict are statistically superior to the available Focus at the 10% significance level, with these two tying with the predictive performance of the ex-post Focus. For the 3-month forecast, the hybrid Phillips curve for aggregate inflation is subtly superior to the Focus in numerical terms but without statistical significance. For the other horizons, no model performs better than the survey-based expectations. There are some potential explanations for this poor performance of the models. First, since we are analyzing the first sub-period, a small sample may have affected the estimates, contributing to the models' poor forecast performance. Second, the instabilities in the Brazilian economy in 2014 and 2015 that resulted in a sharp increase in inflation in 2015, as well as the rapid disinflation that occurred from the second half of 2016, are challenging events to anticipate, especially without extensive historical data. Conversely, from 2017 until the beginning of the COVID-19 pandemic, Brazilian inflation remained reasonably controlled and close to the inflation target, leaving limited opportunities for models to enhance survey-based expectations. These dynamics of the Brazilian inflation can be observed from Figure 3 later on.

Since the COVID-19 pandemic, most models perform statistically better than the Focus survey, as seen in Table 3. Models that look directly at aggregate inflation or use some inflation disaggregation tend to perform very well at all horizons, including nowcasting. Specifically, the adaLASSO, factor model, and FarmPredict working directly with the aggregate inflation achieve RMSE 10% lower than the RMSE of the available Focus and smaller RMSE than the ex-post Focus, something challenging to imagine before the pandemic. Meanwhile, augmented AR and ridge, both employing BCB disaggregation, also perform well. One month ahead, even some models employing the highest level of disaggregation (i.e., subgroups) deliver good results. From the results in this second sub-period, we understand the good performance of the RF for the entire period. Considering aggregate inflation, the method already registers the best performances from $h = 3$, with similar results when using any disaggregation. Frequently, the RF can reduce the RMSE of the available and ex-post Focus by up to 30% for longer horizons, regardless of whether considering aggregate inflation or some disaggregation. For 12-month accumulated inflation, the RF obtains the best performance when using the BCB disaggregation: a 35% reduction compared to the RMSE of the main benchmarks. Finally, regarding model combinations, each is statistically superior to Focus for all $h \geq 1$, often at a 1% significance

Table 2: Out-of-sample RMSE with respect to the available Focus: Jan/2014 to Feb/2020

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	$\Sigma 12m$
A. Survey													
Focus (available)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Focus (<i>ex-post</i>)	0.932***	0.969***	0.995**	1.001	0.999	1.001	0.999	1.000	1.000	1.000	0.999	1.002	0.996***
B. Aggregate inflation													
RW	3.088	1.752	1.428	1.474	1.574	1.547	1.516	1.427	1.260	1.240	1.307	1.382	2.114
Historical Mean	3.309	1.492	1.246	1.208	1.184	1.173	1.155	1.146	1.133	1.125	1.127	1.134	1.422
AR	2.781	1.466	1.225	1.240	1.266	1.244	1.172	1.122	1.093	1.098	1.121	1.155	1.339
HNKPC	0.966	1.008	1.022	0.994	1.083	1.087	1.061	1.043	1.046	1.050	1.072	1.079	1.089
Augmented AR	1.015	1.067	1.081	1.090	1.114	1.101	1.082	1.076	1.068	1.086	1.117	1.126	1.161
adaLASSO	0.990	0.979	1.009	1.052	1.031	1.074	1.086	1.084	1.134	1.136	1.176	1.097	1.136
Factor	0.990	0.970*	1.001	1.016	1.056	1.055	1.056	1.043	1.106	1.121	1.113	1.115	1.109
FarmPredict	0.990	0.970*	1.003	1.024	1.065	1.040	1.062	1.094	1.113	1.129	1.105	1.163	1.102
Target Factor	1.499	1.225	1.122	1.338	1.334	1.215	1.110	1.223	1.212	1.231	1.171	1.118	1.333
CSR	0.968	1.011	1.035	1.081	1.139	1.104	1.086	1.063	1.056	1.047	1.043	1.054	1.113
Random Forest	1.918	1.276	1.156	1.151	1.137	1.144	1.130	1.130	1.101	1.113	1.110	1.127	1.319
C. Disaggregation: tradable, nontradable and monitored prices (BCB)													
AR	2.943	1.535	1.268	1.275	1.304	1.318	1.241	1.261	1.150	1.086	1.079	1.077	1.519
Augmented AR	1.049	1.120	1.131	1.150	1.182	1.189	1.164	1.186	1.137	1.163	1.189	1.153	1.335
Ridge	1.100	1.190	1.134	1.129	1.167	1.242	1.125	1.111	1.088	1.069	1.144	1.060	1.363
adaLASSO	1.225	1.098	1.060	1.092	1.112	1.049	1.064	1.069	1.110	1.095	1.106	1.062	1.168
Factor	1.094	1.098	1.120	1.142	1.142	1.082	1.075	1.074	1.089	1.084	1.098	1.101	1.178
FarmPredict	1.226	1.118	1.095	1.116	1.110	1.072	1.072	1.050	1.092	1.107	1.101	1.085	1.180
Target Factor	1.569	1.183	1.078	1.221	1.372	1.260	1.127	1.215	1.123	1.239	1.175	1.118	1.309
CSR	2.364	1.401	1.221	1.188	1.196	1.162	1.168	1.184	1.133	1.123	1.153	1.105	1.362
Random Forest	2.223	1.322	1.161	1.167	1.163	1.182	1.164	1.161	1.131	1.107	1.112	1.134	1.380
D. Disaggregation: groups (IBGE)													
AR	3.236	1.659	1.301	1.319	1.272	1.199	1.176	1.221	1.236	1.268	1.237	1.314	1.465
Augmented AR	1.084	1.305	1.270	1.263	1.276	1.238	1.199	1.195	1.177	1.261	1.269	1.346	1.380
Ridge	2.043	1.443	1.205	1.171	1.149	1.136	1.111	1.107	1.084	1.098	1.106	1.116	1.329
adaLASSO	1.361	1.273	1.159	1.146	1.128	1.110	1.154	1.155	1.163	1.186	1.148	1.146	1.302
Factor	1.411	1.270	1.164	1.163	1.157	1.157	1.127	1.105	1.105	1.114	1.109	1.119	1.289
FarmPredict	1.510	1.226	1.148	1.174	1.181	1.138	1.108	1.113	1.089	1.139	1.128	1.149	1.322
Target Factor	1.553	1.323	1.227	1.315	1.215	1.172	1.166	1.181	1.244	1.167	1.210	1.185	1.358
CSR	2.410	1.434	1.318	1.285	1.231	1.159	1.083	1.074	1.171	1.148	1.138	1.111	1.248
Random Forest	2.282	1.342	1.174	1.216	1.185	1.170	1.154	1.170	1.167	1.146	1.120	1.127	1.409
E. Disaggregation: subgroups (IBGE)													
AR	3.467	1.745	1.461	1.503	1.503	1.464	1.448	1.538	1.510	1.501	1.453	1.665	1.788
Augmented AR	1.345	1.529	1.547	1.547	1.545	1.496	1.418	1.462	1.498	1.494	1.476	1.745	1.634
Ridge	3.208	1.448	1.214	1.185	1.151	1.142	1.124	1.107	1.108	1.111	1.110	1.118	1.390
adaLASSO	1.533	1.261	1.264	1.174	1.141	1.117	1.105	1.166	1.149	1.150	1.200	1.166	1.308
Factor	1.521	1.266	1.180	1.149	1.156	1.133	1.108	1.106	1.104	1.110	1.137	1.114	1.280
FarmPredict	1.541	1.273	1.136	1.148	1.146	1.121	1.117	1.098	1.081	1.143	1.176	1.173	1.302
Target Factor	1.450	1.296	1.337	1.278	1.280	1.184	1.141	1.153	1.255	1.171	1.218	1.244	1.407
CSR	2.457	1.394	1.319	1.244	1.201	1.155	1.096	1.107	1.104	1.138	1.177	1.119	1.253
Random Forest	2.325	1.360	1.180	1.224	1.194	1.199	1.182	1.168	1.150	1.117	1.106	1.117	1.407
F. Model combinations for disaggregates													
Aggreg. Comb.	1.273	1.077	1.024	1.042	1.076	1.059	1.048	1.046	1.050	1.058	1.075	1.083	1.142
BCB Comb.	1.393	1.168	1.097	1.119	1.138	1.122	1.096	1.107	1.084	1.086	1.093	1.066	1.275
Groups Comb.	1.610	1.271	1.142	1.158	1.133	1.100	1.083	1.090	1.104	1.117	1.107	1.126	1.290
Subgroups Comb.	1.739	1.290	1.187	1.180	1.163	1.138	1.105	1.118	1.133	1.130	1.143	1.172	1.341

Notes: see Table 1.

level. However, the combinations do not beat some individual models with good predictive performance in the sub-period.

Figure 3 presents the temporal evolution of actual inflation, Focus survey expectations, and the best aggregate- and disaggregates-based models for each horizon. Looking at the projections for $h = 0$, we partially understand why it is not easy to outperform the survey in the very-short term. The Focus consensus is very close to the actual values. Furthermore, as we will see in Subsection 5.3, available inflation expectations are the primary predictor for model-based nowcasting. The survey contains much relevant information unavailable to the econometrician, so we already expect this result. When analyzing the other horizons ($h \geq 1$), we note that it is challenging for the survey and models to predict peaks and valleys of inflation. Already at $h = 1$, we observe outstanding

Table 3: Out-of-sample RMSE with respect to the available Focus: Mar/2020 to Jun/2022

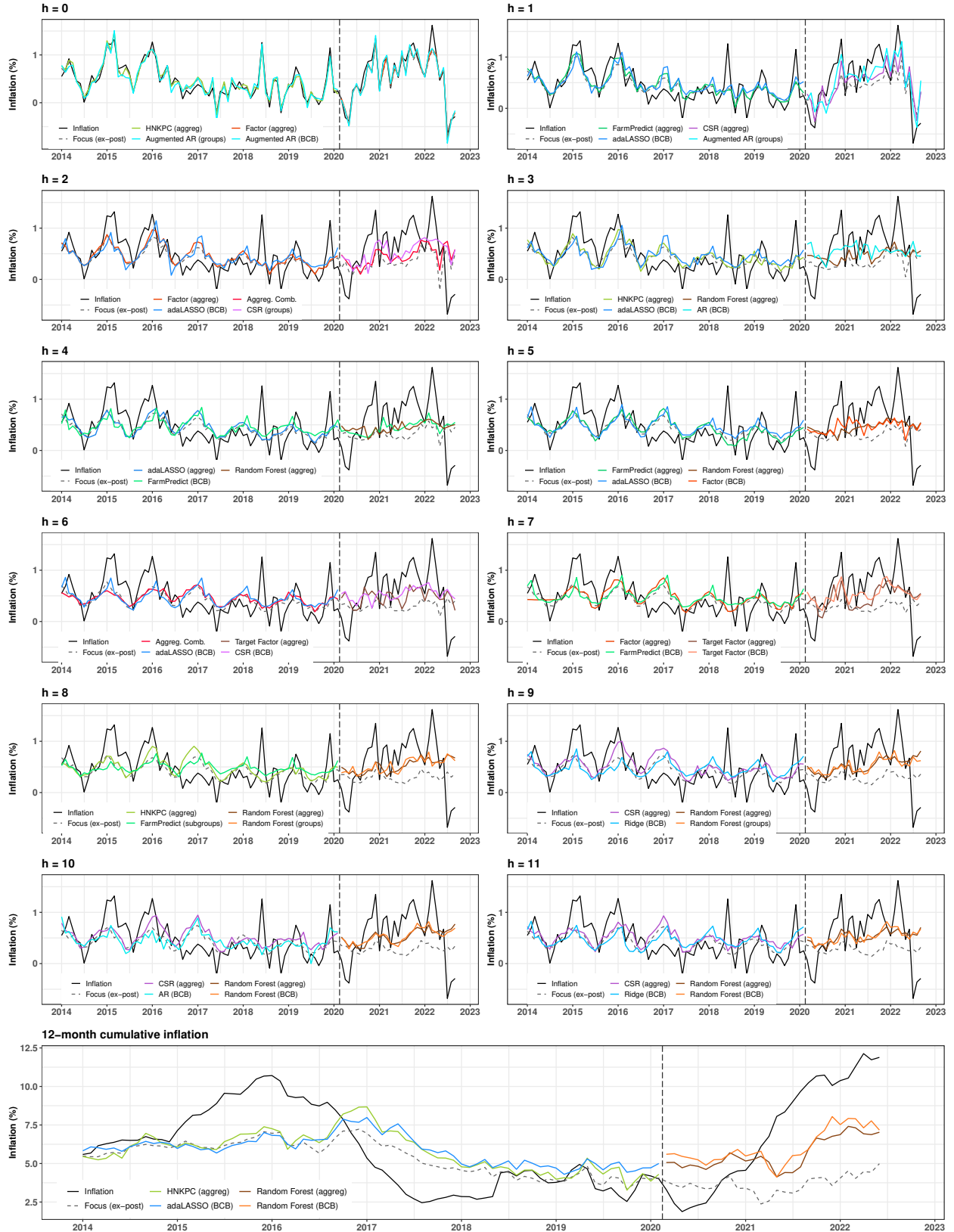
Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	$\Sigma 12m$
A. Survey													
Focus (available)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Focus (<i>ex-post</i>)	0.934**	0.975***	0.991***	1.001	1.000	0.999	1.001	0.999	0.999	1.000	1.002	1.001	1.000
B. Aggregate inflation													
RW	2.443	1.189	1.023	1.056	1.003	1.024	0.983	0.965	0.939	0.908	0.920	0.920	1.128
Historical Mean	2.915	1.203	0.948	0.921**	0.906**	0.888***	0.868***	0.862***	0.863***	0.862***	0.860***	0.857***	0.835***
AR	2.264	1.060	0.904*	0.938*	0.937*	0.910**	0.871***	0.827***	0.807***	0.803***	0.828***	0.846***	0.799***
HNKPC	0.927*	0.913***	0.933***	0.971	0.999	0.986	0.917***	0.911**	0.896***	0.913***	0.940***	0.976*	0.938***
Augmented AR	0.924*	0.904**	0.915**	0.951	0.949*	0.916**	0.868***	0.834***	0.825***	0.856***	0.889***	0.927**	0.854***
adaLASSO	0.896**	0.922**	0.954***	0.948***	0.935***	0.921***	0.889***	0.890***	0.868***	0.893***	0.933***	0.908***	0.879***
Factor	0.894**	0.896***	0.939***	0.923***	0.907***	0.921***	0.938***	0.943***	0.944***	0.931***	1.005	0.990	0.935***
FarmPredict	0.896**	0.918***	0.929***	0.958**	0.947**	0.971	0.959**	0.964**	0.947**	0.953**	0.965*	0.962*	0.956**
Target Factor	1.018	1.043	0.986	0.998	0.934	0.948	0.789***	0.778***	0.788***	0.740**	0.889*	1.089	0.772***
CSR	0.954	0.851***	0.904***	0.903**	0.925**	0.909**	0.875***	0.840***	0.845***	0.917**	1.041	1.083	0.851***
Random Forest	1.370	0.951	0.916*	0.896***	0.877***	0.855***	0.815***	0.794***	0.761***	0.740***	0.723***	0.710***	0.689***
C. Disaggregation: tradable, nontradable and monitored prices (BCB)													
AR	2.161	0.931	0.831**	0.861**	0.895*	0.876**	0.815***	0.841***	0.853***	0.910*	0.985	1.006	0.779***
Augmented AR	0.903*	0.841**	0.914**	0.991	1.024	0.977	0.883***	0.863***	0.849***	0.865***	0.938	1.011	0.904***
Ridge	0.930*	0.842**	0.903**	0.929**	0.944**	0.950*	0.898***	0.870***	0.871***	0.870***	0.904***	0.919***	0.859***
adaLASSO	0.959	0.907*	0.895***	0.866***	0.864***	0.867***	0.881***	0.868***	0.869***	0.854***	0.859***	0.873***	0.769***
Factor	0.948	0.886***	0.884***	0.861***	0.832***	0.808***	0.829***	0.901***	0.928***	0.946***	0.937***	0.938***	0.827***
FarmPredict	1.024	0.940**	0.928***	0.867***	0.849***	0.838***	0.863***	0.934***	0.932***	0.929***	0.933***	0.916***	0.858***
Target Factor	1.046	0.931	0.893**	0.982	0.910	0.924	0.784***	0.727***	0.827***	0.754***	0.881*	1.046	0.729***
CSR	1.973	0.954	0.834***	0.864**	0.860**	0.827***	0.769***	0.796***	0.808***	0.831***	0.912**	1.011	0.719***
Random Forest	1.600	1.042	0.902*	0.955	0.886***	0.845***	0.814***	0.790***	0.748***	0.733***	0.707***	0.716***	0.648***
D. Disaggregation: groups (IBGE)													
AR	2.118	0.888	0.814*	0.825**	0.856**	0.890**	0.981	0.979	1.022	1.043	0.995	0.903**	0.850**
Augmented AR	0.918	0.767**	0.836**	0.871*	0.924*	0.878**	0.960	0.961	0.983	0.973	1.024	1.043	0.841***
Ridge	1.153	1.143	0.907**	0.916**	0.904**	0.907***	0.884***	0.894***	0.881***	0.887***	0.893***	0.905***	0.835***
adaLASSO	1.070	0.910*	0.915**	0.901**	0.898***	0.900***	0.909**	0.849***	0.803***	0.844***	0.851***	0.879***	0.767***
Factor	0.993	0.880**	0.926**	0.922**	0.934**	0.909***	0.942**	0.854***	0.866***	0.946**	0.911***	0.885***	0.861***
FarmPredict	1.134	0.960	0.923**	0.946*	0.922**	0.916**	0.957*	0.893***	0.874***	0.938**	0.888***	0.871***	0.865***
Target Factor	0.904	0.831**	0.934	0.995	0.901*	0.846**	0.812***	0.765***	0.779***	0.792***	0.865**	1.019	0.768***
CSR	1.849	0.831*	0.792***	0.828***	0.789***	0.825***	0.823***	0.802***	0.799***	0.834***	0.846***	0.882***	0.710***
Random Forest	1.895	1.102	0.902*	0.902**	0.855**	0.851***	0.815***	0.760***	0.738***	0.731***	0.737***	0.735***	0.685***
E. Disaggregation: subgroups (IBGE)													
AR	2.715	1.264	0.931	0.939	1.008	0.927	1.009	0.999	1.047	1.158	1.126	1.089	1.067
Augmented AR	1.030	0.982	1.060	1.056	1.061	0.905	0.939	0.995	1.006	1.153	1.180	1.168	0.980
Ridge	2.990	1.179	0.927*	0.927**	0.915**	0.931**	0.916***	0.920**	0.912**	0.896***	0.901***	0.934**	0.903***
adaLASSO	1.210	0.992	0.954	0.960	0.938**	0.925**	0.898***	0.841***	0.841***	0.893***	0.922***	0.936***	0.849***
Factor	1.189	1.037	0.977	0.971	0.943**	0.946**	0.932***	0.887***	0.909***	0.961*	0.962**	0.918***	0.914***
FarmPredict	1.209	1.034	0.975	0.951	0.930**	0.916**	0.925***	0.886***	0.903***	0.942**	0.965**	0.935***	0.904***
Target Factor	0.964	0.810*	0.979	0.998	0.977	0.959	0.838**	0.744***	0.823***	0.807**	0.908	1.012	0.810***
CSR	1.963	1.033	0.853***	0.914**	0.875***	0.923***	0.882***	0.829***	0.795***	0.857***	0.913***	0.961	0.832***
Random Forest	1.865	1.114	0.908*	0.926*	0.869**	0.870**	0.825***	0.781***	0.768***	0.750***	0.746***	0.746***	0.725***
F. Model combinations for disaggregates													
Aggreg. Comb.	1.127	0.909***	0.895***	0.918***	0.906***	0.898***	0.858***	0.845***	0.840***	0.840***	0.886***	0.910***	0.835***
BCB Comb.	1.075	0.866***	0.862***	0.888***	0.877***	0.856***	0.820***	0.827***	0.838***	0.839***	0.878***	0.923***	0.777***
Groups Comb.	1.159	0.864**	0.833***	0.866***	0.864***	0.861***	0.880***	0.842***	0.836***	0.868***	0.869***	0.888***	0.786***
Subgroups Comb.	1.411	0.975	0.906**	0.932**	0.921***	0.899***	0.882***	0.840***	0.860***	0.898**	0.929**	0.944**	0.869***

Notes: see Table 1.

forecasting errors. One consequence of COVID-19, which start is highlighted by vertical dashed lines on each plot, is that the Focus survey initially overestimated inflation and afterward systematically underestimated one. Despite the challenges of generating accurate forecasts in such an uncertain period, several models perform better than expert forecasts for all horizons. A punctual example is the adaLASSO that, using both aggregates and BCB disaggregations, as well as other models, achieves a great result in forecasting the peak observed in December 2020 at $h = 1$, a point at which the available inflation expectation is far from the actual value. Thus, other variables besides the available inflation expectations are fundamental for the performance of model-based forecasts.

Remarks. The good performance of the RF is mainly due to its ability to capture the higher level of future inflation from the second half of 2020. The model generates forecasts closer to the actual inflation than the Focus expectations, which systematically underestimate inflation in that period.

Figure 3: Forecasts by each horizon and 12-month cumulative period



Notes: Black solid lines indicate the actual inflation. Gray dashed lines indicate the median of the ex-post inflation expectations from the Focus survey on the last business day of each month. Solid-colored lines indicate forecasts generated by different models or a combination of models. The vertical dashed lines separate the period before and after the COVID-19 pandemic.

Since the pandemic, models for disaggregated inflation tend to provide more accurate forecasts than models for aggregate inflation, except for nowcasting. For each $h \geq 2$, we note that adaLASSO, factor model, FarmPredict, and CSR using any disaggregation of inflation deliver forecasts with lower RMSE than the respective models using aggregate inflation, with a few exceptions for the CSR using groups and subgroups that do not outperform the CSR using aggregate inflation. In turn, the RF performs well regardless of the target variable. These findings underscore the use of models in inflation forecasting, including the junction between disaggregated analysis and machine learning techniques, particularly during periods of higher economic instability, such as a pandemic. Previous studies such as [Altug and Çakmaklı \(2016\)](#) and [Medeiros, Vasconcelos, Veiga, and Zilberman \(2021\)](#) have also shown that models perform well during more volatile periods. Next, we will analyze the forecasts for disaggregates and identify the predictors selected by adaLASSO and FarmPredict, methods that allow variable selection.

5.3 Forecast of disaggregates and variable selection

5.3.1 Disaggregation into BCB categories

Predictive performance. Now we consider the predictive performance of the models using different disaggregations, starting with the disaggregation from the BCB. Since we lack long time series for survey-based expectations for disaggregated inflation, we use the AR model as a benchmark. Of all the BCB disaggregations, monitored prices are the most challenging to forecast since they are subject to many unexpected changes resulting from government decisions that often do not freely follow supply and demand movements. According to the results displayed in Table 4, other models manage to beat the AR only in short or more distant horizons for this disaggregate (Panel A). In nowcasting, other methods perform significantly better than the AR, with augmented AR and Ridge standing out by obtaining more than a 30% reduction in RMSE. For $h = 1$, augmented AR achieves a 10% reduction in RMSE, the only statistically significant at any level. Some models present minor RMSE for intermediate horizons, but the results are not statistically significant according to the DM test. For ten and eleven months ahead, Ridge delivers reductions of 4% and 6% in RMSE, respectively, compared to the AR, and, specifically for $h = 11$, adaLASSO, factor model, and FarmPredict also statistically outperform the AR, with RMSE reductions ranging from 3% and 4%. Putting all horizons together, Ridge, adaLASSO, FarmPredict, and factor model generate more accurate forecasts than the AR by delivering RMSEs 2% to 4% lower than the AR model, all statistically significant at the 1% level. The good result of the augmented AR model is restricted to the short term.

Looking at the other BCB disaggregates, namely non-tradable and tradable items, we notice that machine learning models deliver better results than the traditional AR model (Panels B and C of Table 4). For non-tradables, once again augmented AR and Ridge stand out in nowcasting with RMSE reductions of 18% and 16%, while adaLASSO and FarmPredict achieve the best performances between one and five months ahead, with RMSE reductions oscillating between 20% and 27%. For all $h \geq 5$, the random forest dominates by delivering the lowest RMSE. Aggregating all horizons, all ML models perform statistically better than AR for non-tradable items. In turn, the results for tradables are similar, with the ML methods yielding subtly smaller improvements. Also for tradables, there is a predominance of the RF: this method obtains the best or second-best performance for all $h \geq 1$, with RMSE reductions ranging from 10% to 19%. Other methods that stand out are adaLASSO, which obtains the best result in nowcasting (almost 24% reduction in RMSE), and target factor, which registers

Table 4: Out-of-sample RMSE for BCB disaggregate (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate and horizon

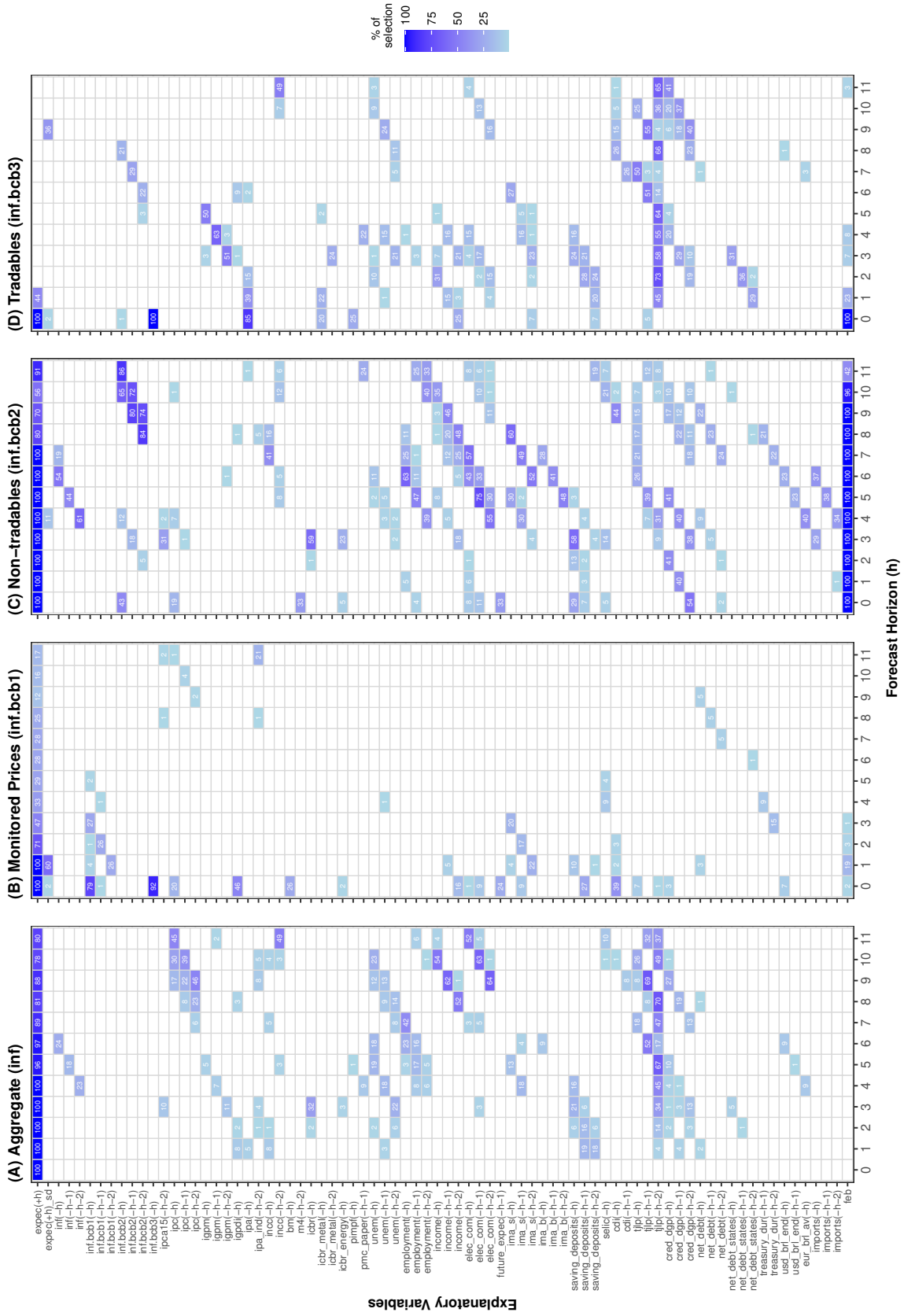
Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
A. Monitored Prices													
AR	1.000	1.000	1.000	1.000	1.000	1.000	<i>1.000</i>	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.694 ***	0.903 ***	1.015	1.022	1.025	1.023	1.026	1.035	1.038	1.037	1.021	1.033	0.996
Ridge	<i>0.709</i> ***	<i>0.954</i>	<i>0.987</i>	0.984	0.984	<i>0.988</i>	1.001	<i>0.985</i>	0.988	0.986	0.957 **	0.943 **	0.960 ***
adaLASSO	0.756***	0.960	0.986	<i>0.988</i>	<i>0.987</i>	0.987	1.004	0.989	<i>0.996</i>	1.004	0.971	0.962*	<i>0.969</i> ***
Factor	0.734***	0.970	1.008	1.003	0.999	0.999	1.016	1.009	1.006	<i>0.996</i>	0.976	0.961*	0.977***
FarmPredict	0.774***	0.980	1.002	0.998	0.997	1.009	0.998	0.982	0.996	0.998	0.974	0.965*	0.975***
Target Factor	0.750***	1.033	1.045	1.082	1.063	1.079	1.086	1.149	1.143	1.097	1.073	1.080	1.063
CSR	0.974	1.002	1.022	1.034	1.061	1.053	1.034	1.046	1.066	1.057	1.021	1.027	1.034
Random Forest	0.872***	1.020	1.039	1.060	1.058	1.076	1.098	1.068	1.061	1.033	<i>0.967</i>	<i>0.956</i>	1.027
B. Non-Tradables													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	<i>0.843</i> **	0.831***	0.844***	0.820***	0.810***	0.787***	0.829***	0.853**	0.907**	0.963	0.985	0.975	0.869***
Ridge	0.824 ***	0.817***	0.830***	0.808***	0.796***	0.780***	0.823***	<i>0.847</i> ***	<i>0.900</i> **	<i>0.944</i>	0.957	0.950	0.854***
adaLASSO	0.896*	0.784 ***	0.778 ***	<i>0.777</i> ***	<i>0.748</i> ***	<i>0.733</i> ***	<i>0.812</i> ***	0.850**	0.944	0.977	0.955	0.962	<i>0.847</i> ***
Factor	0.889**	0.809***	0.806***	0.804***	0.765***	0.774***	0.823***	0.866**	0.951	1.006	0.968	0.950	0.864***
FarmPredict	0.895*	<i>0.798</i> ***	<i>0.782</i> ***	0.768 ***	0.744 ***	0.739***	0.842***	0.855**	0.955	1.000	0.968	<i>0.938</i>	0.853***
Target Factor	1.040	0.846**	0.896	1.022	0.935	0.799***	0.885*	1.030	1.026	1.015	<i>0.944</i>	1.045	0.954**
CSR	0.892***	0.864***	0.887***	0.887***	0.867***	0.816***	0.856***	0.875***	0.931*	0.956	0.968	1.013	0.899***
Random Forest	0.868***	0.855***	0.816***	0.803***	0.778***	0.729 ***	0.791 ***	0.805 ***	0.854 ***	0.886 **	0.893 **	0.918 *	0.829 ***
C. Tradables													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.832**	0.911**	0.995	1.070	1.082	1.051	0.998	0.948**	0.951*	0.989	1.044	1.055	1.003
Ridge	0.810***	0.920**	0.954**	0.982	0.960**	1.042	0.899***	0.874***	0.917**	0.935**	1.030	0.889***	0.940***
adaLASSO	0.763 ***	0.941*	0.943**	<i>0.940</i> ***	<i>0.909</i> ***	<i>0.844</i> ***	0.848***	0.861***	0.918**	0.886***	<i>0.912</i> **	<i>0.834</i> ***	<i>0.886</i> ***
Factor	0.814***	0.928**	0.972	0.956*	0.927**	0.860***	0.845***	0.882***	0.932**	0.966	0.937*	0.906***	0.913***
FarmPredict	0.811***	0.967	0.980	0.967	0.925**	0.872***	0.858***	0.897***	0.940*	0.956*	0.937*	0.899***	0.920***
Target Factor	<i>0.792</i> **	0.886 **	<i>0.907</i> **	0.979	1.000	0.935	<i>0.821</i> ***	0.769 ***	0.831 ***	0.896**	0.990	0.926*	0.900***
CSR	0.884**	1.045	1.079	1.003	0.917*	0.886***	0.909***	0.884***	0.866***	<i>0.873</i> ***	0.925**	0.932***	0.933***
Random Forest	0.820***	<i>0.903</i> ***	0.906 ***	0.902 ***	0.853 ***	0.824 ***	0.819 ***	<i>0.810</i> ***	<i>0.836</i> ***	0.848 ***	0.841 ***	0.810 ***	0.847 ***

Notes: ***, **, and * indicate that for a specific disaggregate and forecast horizon, a model m performed statistically better than an AR model at 1, 5, and 10% significance levels in a one-tailed Diebold-Mariano test with $\mathbb{H}_0 : \text{MSE}(\hat{\pi}_{i,t+h|t}^m) = \text{MSE}(\pi_{i,t+h|t}^{\text{AR}})$ versus $\mathbb{H}_1 : \text{MSE}(\hat{\pi}_{i,t+h|t}^m) < \text{MSE}(\pi_{i,t+h|t}^{\text{AR}})$. The value highlighted in bold blue indicates the best model for each horizon in terms of RMSE ratio with respect to the AR model, and the values in blue italics indicate the second and third best models. The average weights of each disaggregate in IPCA are: monitored prices (25%); non-tradables (41.5%); and tradables (33.5%).

the best performance one, seven, and eight months ahead. By gathering the forecasts for all periods, the RF obtains an average reduction of 15% in RMSE compared to the AR model, with adaLASSO coming close behind, with a reduction of 11% in RMSE. All models, except the augmented AR, outperform the AR at the 1% significance level. These findings suggest that models that include more predictors, impose restrictions on parameters, or assume other functional forms can be more advantageous in inflation forecasting than traditional time-series models such as the AR model. Lastly, we note that the improvements due to ML methods in a data-rich environment are not just observed for the pandemic period, as seen in Tables C1 and C2 in Appendix C.

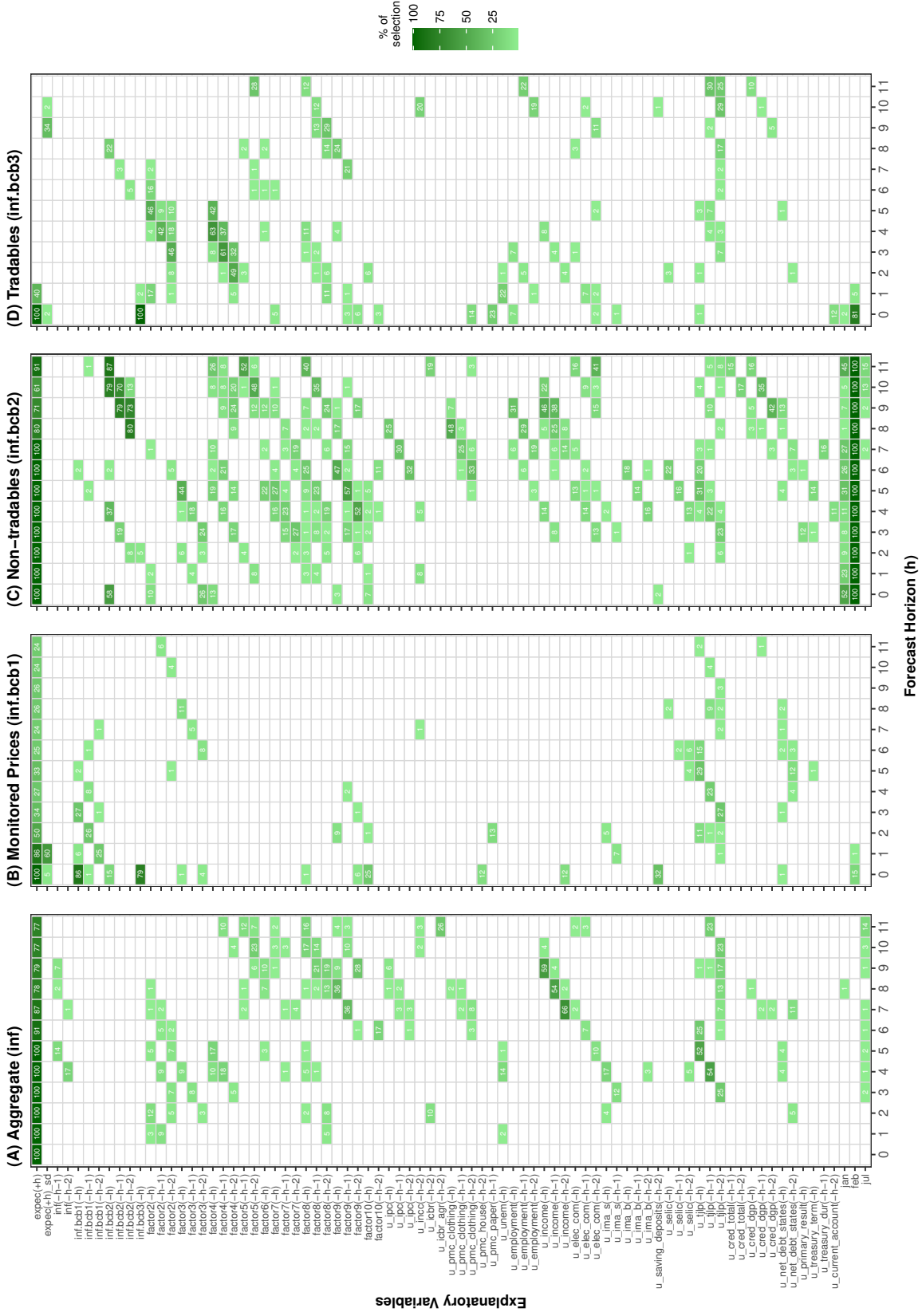
Variable selection. To explore potential economic intuitions for the results from aggregate inflation and BCB disaggregation, we compare what is behind the two approaches in terms of variable selections by adaLASSO and FarmPredict, as shown in Figures 4 and 5, respectively. In both Figures, panel A brings the predictors that each method selected to forecast the aggregate inflation directly. Meanwhile, panels B, C, and D display the predictors selected to predict price variation of administered, non-tradable, and tradable items. To make the presentation viable, we restricted ourselves to the variables chosen at least 20% and 11.5% of the time in at least one forecast horizon, respectively. Variables definitions are shown in Table B1 in Appendix B. The prefix “u_” in some variables in Figure 5 indicates that the variable had the common factors “discounted” and, therefore, only its idiosyncratic component is left. These variables are indicated by $\hat{u}_j$ in Equation (3.3).

Figure 4: adaLASSO selection: aggregate inflation and BCB disaggregates (% of extending windows)



Notes: We cut out variables that do not exceed 20% of selection at least one forecast horizon. The definitions of the variables are in Table B1.

Figure 5: FarmPredict selection: aggregate inflation and BCB disaggregates (% of extending windows)



Notes: We cut out variables that do not exceed 11.5% of selection at least one forecast horizon. The definitions of the variables are in Table B1.

We can summarize the results of the variable selections in the following topics:

1. *High variability in the type of selected predictors.* The adaLASSO selects all ten classes of predictors (see Table B1 in Appendix B). Even controlling for common factors (FarmPredict), each class of predictors still appears. This result shows the importance of considering a broad set of information.
2. *Low participation of economic activity variables.* Variables related to economic activity appear little when we control for common factors via FarmPredict and almost never for adaLASSO. There exists some correlation between activity variables and monetary base, M1 and M2 money supplies, variables occasionally select, for example. Beyond that, the information linked to economic activity variables may be contained in other relevant variables (e.g., inflation expectations), including the possibility of a non-linear relationship between these variables. Thus, we must be careful not to conclude that economic activity variables are irrelevant for forecasting inflation.
3. *Available inflation expectations (survey) are frequently picked.* adaLASSO and FarmPredict for aggregate and non-tradable items frequently select the inflation expectation (expec) at all horizons. For administrated and tradable items, the selection of the expectation decreases as the horizon increases, appearing only in the very short term for tradables. We note that the expectation is about aggregate inflation, so it is reasonable that it is not relevant to explain some specific disaggregate. Furthermore, due to their greater share, non-tradables present a more remarkable similarity with aggregate inflation than the other breakdowns.
4. *Prices (including commodities) are often chosen.* According to the variable selections, Brazilian inflation indexes and commodities price variations are relevant to forecast official Brazilian inflation. Various indexes and price variations carry relevant information to forecast Brazilian inflation. In addition, due to past inflationary history, Brazil has several monthly indexes calculated by different organizations that cover different periods (e.g., days 1 to 30, 11 to 10, and 21 to 30). Thus, we must consider this information when forecasting Brazilian inflation.
5. *Factors that explain most of the variability of predictors are not always more relevant to forecasting inflation.* Interestingly, the common factor that explains most of the variability of the predictors (i.e., factor1) is rarely select to forecast aggregate inflation or any disaggregate. Instead, the following factors up to the tenth are chosen for various horizons, mainly to predict non-tradables, lesser extent for aggregate and tradables, and very little for administrated ones.
6. *Non-tradables record the richer structure of predictors; monitored items, the poorer.* The good predictability of non-tradables is potentially related to the larger number of predictors. Note that non-tradables have more predictors than aggregate inflation itself. This finding underscores the importance of looking at disaggregates. For example, the February dummy's relevance for forecasting inflation only appears when we consider the forecast of non-tradable inflation. This dummy is crucial because it captures the variation in education prices, a sector whose contracts are usually updated in January. In turn, the price variation of monitored items has few predictors, contributing to this disaggregate being the most challenging for forecasting.

5.3.2 Disaggregation into IBGE groups

Now we address the predictive performance for each IBGE group. The identification of each group, as well as their respective participation in the IPCA, are available in Table A1 in Appendix A. From Table 5, we note that ML methods perform statistically better than the AR model and numerically better than the augmented AR for all components by stacking the horizons. An exception is the target factor, which does not perform well for some groups. Furthermore, from Table C3 in Appendix C, which presents detailed results of each disaggregate by forecast horizon, we notice that there are infrequent horizons that do not have an ML model performing statistically better than the benchmark.

Table 5: Out-of-sample RMSE for IBGE groups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate, joining all horizons

Estimator/Model	inf.g1	inf.g2	inf.g3	inf.g4	inf.g5	inf.g6	inf.g7	inf.g8	inf.g9
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.007	0.977**	1.062	0.874***	1.001	0.881***	1.006	0.451***	1.066
Ridge	0.886***	0.908***	0.956***	0.836***	0.966**	0.889***	0.917***	0.446***	0.805***
adaLASSO	0.900***	0.913***	0.905***	0.844***	0.961***	0.995	0.886***	0.457***	0.790***
Factor	0.911***	0.915***	0.900***	0.863***	0.959***	0.994	0.876***	0.450***	0.795***
FarmPredict	0.909***	0.919***	0.912***	0.920***	0.966***	1.015	0.882***	0.457***	0.798***
Target Factor	0.932***	0.992	0.905***	0.884***	1.026	0.909***	1.008	0.487***	1.164
CSR	0.954**	0.949***	0.849***	0.898***	0.951***	0.974*	0.927***	0.698***	0.863***
Random Forest	0.844***	0.961***	0.845***	0.866***	0.968***	0.945***	0.931***	0.573***	0.838***

Notes: see Table 4. For a definition of the groups and their respective weights in the IPCA, see Table A1 in Appendix A.

Different models perform better for different disaggregates. Education (inf.g8) is the group most benefited from using other techniques. However, given the good performance of the augmented AR, we infer that predictive improvement is mainly due to the inclusion of the February dummy. Except for augmented AR and target factor, the models also achieve predictive improvement for communication (inf.g9), including all horizons individually. Transportation is the group for which the models beat the AR model by stacking the horizons with the smallest margin (inf.g5). In addition, the ML methods are not statistically superior to AR in half of the forecast horizons for this disaggregate. The transportation group comprises public transport fares and expenses with own vehicle and fuel, mostly items whose prices are administered by the government, which are difficult to forecast. However, except once again for augmented AR and target factor, the models deliver a statistically significant average reduction of at least 3% in RMSE compared to the AR model, with Ridge’s predictive gain around 10% between 6- and 9-month-ahead. Lastly, it is worth highlighting the good performance of the RF to forecast the price variation of foods and beverages (inf.g1) at all horizons.

5.3.3 Disaggregation into IBGE subgroups

Lastly, we examine the predictive performance for each IBGE subgroup. The results stacking all horizons are shown in Table 6, and results for each horizon are in Table C4 in Appendix C. Descriptions of subgroups are in Table A1 in Appendix A. Similar to what happens with the disaggregation into groups, ML models achieve more accurate forecasts in comparison to AR models for each subgroup, but with no single model emerging as a dominant predictor for different disaggregates. The reduction of RMSE reaches 71% in the case of courses, reading, and stationery (inf.sg18), 40% for communication (inf.sg19), and 35% for household operations (inf.sg7) and personal services (inf.sg16). RF stout out for delivering the best predictive performances considering all forecast hori-

zons for food at home (*inf.sg1*), appliances (*inf.sg6*), household operations (*inf.sg7*), and fabrics (*inf.sg10*). The Ridge performs well at all horizons for domestic fuels and energy (*inf.sg4*) and jewelry (*inf.sg10*), and the adaLASSO performs well for communication (*inf.sg19*) at all horizons as well. Furthermore, we show outstanding performances for specific horizons: adaLASSO performs well in the long-term for food away from home (*inf.sg2*), target factor in short- and intermediate-term for pharmaceutical and optical products (*inf.sg13*), while the RF performs well at short term for that one and at more distant horizons for the latter.

Table 6: Out-of-sample RMSE for IBGE subgroups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate, joining all horizons

Estimator/Model	inf.sg1	inf.sg2	inf.sg3	inf.sg4	inf.sg5	inf.sg6	inf.sg7	inf.sg8	inf.sg9	inf.sg10
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.062	1.053	1.080	1.000	1.063	1.085	1.050	0.902***	1.040	1.106
Ridge	0.742***	0.798***	0.804***	0.794***	0.854***	0.827***	0.681***	0.886***	0.832***	0.693***
adaLASSO	0.766***	0.759***	0.781***	0.803***	0.823***	0.826***	0.677***	0.737***	0.768***	0.721***
Factor	0.751***	0.803***	0.792***	0.805***	0.838***	0.810***	0.684***	0.759***	0.780***	0.725***
FarmPredict	0.757***	0.795***	0.793***	0.810***	0.839***	0.815***	0.687***	0.809***	0.784***	0.728***
Target Factor	0.786***	0.811***	0.857***	0.954	0.858***	0.836***	0.812***	0.731***	0.802***	0.808***
CSR	0.779***	0.796***	0.785***	0.850***	0.801***	0.819***	0.683***	0.810***	0.791***	0.730***
Random Forest	0.706***	0.772***	0.770***	0.858***	0.849***	0.757***	0.650***	0.759***	0.734***	0.694***
Estimator/Model	inf.sg11	inf.sg12	inf.sg13	inf.sg14	inf.sg15	inf.sg16	inf.sg17	inf.sg18	inf.sg19	
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
Augmented AR	1.147	1.037	0.889***	1.010	1.023	1.117	1.061	0.436***	1.022	
Ridge	0.736***	0.874***	0.941**	0.987	0.899***	0.786***	0.742***	0.415***	0.613***	
adaLASSO	0.706***	0.861***	0.827***	0.979*	0.898***	0.713***	0.737***	0.389***	0.599***	
Factor	0.736***	0.863***	0.892***	0.970**	0.899***	0.719***	0.734***	0.383***	0.605***	
FarmPredict	0.739***	0.861***	0.877***	0.978*	0.896***	0.730***	0.738***	0.389***	0.604***	
Target Factor	0.774***	0.923***	0.729***	0.958**	0.924***	0.807***	0.831***	0.400***	0.829***	
CSR	0.722***	0.867***	0.848***	0.967**	0.897***	0.662***	0.776***	0.604***	0.653***	
Random Forest	0.673***	0.860***	0.798***	0.969**	0.913***	0.643***	0.761***	0.480***	0.639***	

Notes: see Table 4. For a definition of the subgroups and their respective weights in the IPCA, see Table A1 in Appendix A.

Some subgroups are equivalent to groups. It is the case of transportation (*inf.sg12* and *inf.g5*), courses, reading, and stationary (*inf.sg18*) that is equivalent to education (*inf.g8*), and communication (*inf.sg19* and *inf.g9*). Interestingly, the predictive accuracy obtained from disaggregation into subgroups is superior to that based on groups, both stacking the horizons and considering them individually. In particular, the improvements are remarkable for transportation and communication. The difference between both approaches is that, while one includes lags from other subgroups and not groups, the other includes lags from different groups and not subgroups. Note that, stacking the forecast horizons, while CSR reduces the RMSE of the AR model for *group* transportation (*inf.g5*) by 5%, the RF obtains an average reduction of 14% for *subgroup* transportation (*inf.sg12*). For transportation, while the ML methods using disaggregation into groups do not statistically outperform the AR model for horizons 1 to 4, 10, and 11 months ahead, some of these methods employing disaggregation into subgroups generate statistically significant improvements at the most demanding level (i.e., 1%) for all these horizons. For communication, disaggregation into subgroups can improve the already good performance of methods when employing disaggregation into groups.

5.3.4 Further remarks

We find that ML models considering several predictors beat the AR model by a wide and statistically significant margin for all disaggregates considered. Comparing the use of group and subgroup disaggregations, the predictive benefits associated with a higher level of disaggregation appear to

align with the results of [Duarte and Rua \(2007\)](#), for Portugal, [Ibarra \(2012\)](#), for Mexico, and [Bermingham and D’Agostino \(2014\)](#), for the US and Euro Area. However, upon revisiting the results of the aggregation of disaggregated forecasts (Tables 1 to 3), we note that the use of disaggregation into subgroups does not generate more accurate forecasts than the direct approach or aggregation from less profound disaggregations. A potential explanation for this result is that we do not consider the aggregation of disaggregated forecasts generated by different models in this paper. As we have seen, a single model does not dominate the forecasts of all disaggregates and often does not even exhibit dominance over time for the same disaggregate (see Figures from D1 to D4 in Appendix D). Additionally, there is a possibility of inaccurate disaggregated forecasts occurring at some point in time, which can lead to a deterioration of the predictive accuracy of the aggregation of disaggregated forecasts. Importantly, this deterioration cannot be attributed to the use of the most recent available weights for each item in the consumption basket. When we conduct the forecasting exercises again using the actual weights, the results demonstrate no significant changes.

6 Conclusion

In this paper, we investigate the use of inflation disaggregations to forecast the aggregate via aggregation of disaggregated forecasts – what became known as the bottom-up approach. We innovate by considering multi-horizon forecasts of several inflation disaggregates in a data-rich environment (i.e., considering many predictors), which is only possible by employing machine learning methods. Analyzing Brazilian inflation and exploring different levels of disaggregation for inflation, we conduct forecasting exercises from both direct and bottom-up forecast approaches. We highlight the relevance of considering the combination of disaggregated analysis and machine learning methods in the econometrician’s toolbox.

For many forecast horizons, the aggregation of disaggregated forecasts performs as well as survey-based expectations and models that generate forecasts directly from the aggregate. Our results reinforce the benefits of using models in a data-rich environment for inflation forecasting, including aggregating disaggregated forecasts generated from machine learning techniques, mainly during volatile periods. During the COVID-19 pandemic, model-based forecasts, including those based on disaggregated data, tend to provide more accurate forecasts than survey-based expectations. For example, the random forest model based on both aggregate and disaggregated inflation delivers great results for intermediate and longer horizons. The selection of predictors obtained by the adaLASSO and FarmPredict indicates the importance of considering a broad and diversified set of variables when forecasting inflation. Regarding the prediction of individual disaggregates, we find that ML models considering several predictors beat the AR model by a wide and statistically significant margin for all disaggregates and the vast majority of horizons.

This paper can be extended in many ways in future research. Firstly, it is possible to replicate the procedures analyzed here for other developed and emerging countries. Secondly, an important possibility is the formulation and implementation of a methodology that combines different models predicting different disaggregates. As we have seen, there is no dominant technique. Thus, to fully exploit the potential of disaggregated forecasting, it is necessary to combine different forecasts to obtain the aggregate forecast. Beyond combining different models in the dimension of disaggregates, one can also combine different models over horizons to improve the forecast of time-accumulated

inflation (e.g., 12-month cumulative inflation). Thirdly, combining different levels of disaggregation could be valuable since increasing the level in some branches may be advantageous while for others, it does not. For example, in the Brazilian price index employed in this paper, some groups may be worth keeping in the final combination, while others may benefit from further breakdown into subgroups or deeper disaggregation. Finally, a fourth possibility not explored in this paper is to use breakeven inflation as a predictor and benchmark in inflation forecasting.

A Groups and subgroups of the IPCA

Table A1: Description of groups e subgroups of the IPCA

Groups / subgroups	Weight
Foods and beverages	23.7
<i>Food at home</i>	15.8
<i>Food away from home</i>	7.9
Housing	15.5
<i>Utilities and maintenance</i>	10.3
<i>Domestic fuels and energy</i>	5.2
Household goods	4.1
<i>Furniture and fixtures</i>	2.0
<i>Appliances</i>	1.7
<i>Household operations</i>	0.3
Apparel	5.5
<i>Clothes</i>	3.6
<i>Footwear and accessories</i>	1.6
<i>Jewelry</i>	0.3
<i>Fabrics</i>	0.1
Transportation	19.0
<i>Transportation</i>	19.0
Medical and personal care	12.1
<i>Pharmaceutical and optical products</i>	3.6
<i>Medical services</i>	5.5
<i>Personal care</i>	3.0
Personal expenses	10.6
<i>Personal services</i>	6.6
<i>Recreation and tobacco</i>	4.0
Education	5.1
<i>Courses, reading, and stationery</i>	5.1
Communication	4.3
<i>Communication</i>	4.3

Notes: The column “Weight” shows the average weight of each group and subgroup in the IPCA from January 2014 to June 2022. The text indicates each group and subgroup by *inf.g#* and *inf.sg#*, respectively, where the # indicates the order in which each appears in this table.

B Predictive variables

Table B1: Description of predictive variables

#	Abbreviation	Description	Unit	Source	Lag	Transformation
A. Prices and Money						
1	inf, inf.bcb#, inf.g#, inf.sg#	IPCA and disaggregations	index	IBGE	1	% change
2	expec	Focus-based inflation expectation (available)	% per month	BCB	0	–
3	inpc	INPC	index	IBGE	1	% change
4	ipca15	IPCA-15	index	IBGE	0	% change
5	ipc	IPC-Br	index	FGV	1	% change
6	igpm	IGP-M	index	FGV	1	% change
7	igpdi	IGP-DI	index	FGV	1	% change
8	igp10	IGP-10	index	FGV	1	% change
9	ipc.fipe	IPC-Fipe	index	FGV	1	% change
10	ipa	IPA	index	FGV	1	% change
11	ipa_ind	IPA	index	FGV	1	% change
12	ipa_agr	IPA	index	FGV	1	% change
13	incc	INCC	index	FGV	1	% change
14	bm.broad	Broad Monetary Base – end-of-period balance	index	BCB	2	% change
15	bm	Monetary Base – working day balance average	index	BCB	2	% change
16	m1	Money supply M1 – working day balance average	index	BCB	2	% change
17	m2	Money supply M2 – end-of-period balance	index	BCB	2	% change
18	m3	Money supply M3 – end-of-period balance	index	BCB	2	% change
19	m4	Money supply M4 – end-of-period balance	index	BCB	2	% change
B. Commodities Prices						
20	icbr	Brazilian Commodity index (all)	index	BCB	1	% change
21	icbr_agr	Brazilian Commodity index – agriculture	index	BCB	1	% change
22	icbr_metal	Brazilian Commodity index – metal	index	BCB	1	% change
23	icbr_energy	Brazilian Commodity index – energy	index	BCB	1	% change
C. Economic Activity						
24	ibcbr	Brazilian IBC-Br Economic Activity index	index	BCB	3	% change
25	month_gdp	GDP monthly – current prices	R\$ million	BCB	1	% change
26	tcu	Total capacity utilization – manufacturing industry	%	FGV	1	first difference
27	pimpf	Industry Production – general	index	IBGE	2	% change
28	pmc_total	Retail sales volume – total	index	IBGE	2	% change
29	pmc_fuel	Retail sales volume – fuels and oils	index	IBGE	2	% change
30	pmc_supermarket	Retail sales volume – supermarkets and food products	index	IBGE	2	% change
31	pmc_clothing	Retail sales volume – fabrics, clothing and shoes	index	IBGE	2	% change
32	pmc_house	Retail sales volume – furniture and appliances	index	IBGE	2	% change
33	pmc_drugstore	Retail sales volume – pharmaceutical and cosmetic articles	index	IBGE	2	% change
34	pmc_paper	Retail sales volume – books, newspapers and stationery	index	IBGE	2	% change
35	pmc_office	Retail sales volume – office and eletronical equipments	index	IBGE	2	% change
36	pmc_others	Retail sales volume – others	index	IBGE	2	% change
37	pmc_building	Retail sales volume – building material	index	IBGE	2	% change
38	pmc_auto	Retail sales volume – automotive and parts	index	IBGE	2	% change
39	steel	Steel production	index	BCB	1	–
40	prod_vehicles	Vehicle production – total	units	Anfavea	1	% change
41	prod_agr_mach	Production of agricultural machinery – total	units	Anfavea	1	% change
42	vehicle_sales	Vehicle sales by dealerships – total	units	Fenabrave	1	% change
D. Employment						
43	unem	Unemployment (combination of PME and PNADC)	%	IBGE	2	first difference
44	employment	Registered employees by economic activity - Total	units	IBGE	1	% change
45	aggreg_wage	Overall Earnings (broad wage income)	R\$ (million)	BCB	1	% change
46	min_wage	Federal Minimum Wage	R\$	MTb	0	% change
47	income	Households gross disposable national income	R\$ (million)	BCB	2	% change
E. Electricity						
48	elec	Electricity consumption – total	GWh	Eletrobrás	2	% change
49	elec_res	Electricity consumption – residential	GWh	Eletrobrás	2	% change
50	elec_com	Electricity consumption – commercial	GWh	Eletrobrás	2	% change
51	elec_ind	Electricity consumption – industry	GWh	Eletrobrás	2	% change
F. Confidence						
52	cons_confidence	Consumer Confidence index	index	Fecomercio	1	% change
53	future_expec	Future expectations index	index	Fecomercio	1	% change
54	conditions	Current economic conditions index	index	Fecomercio	1	% change

(continued on next page)

Table B1: Description of predictive variables (cont.)

#	Abbreviation	Description	Unit	Source	Lag	Transformation
G. Finance						
55	ibovespa	Ibovespa index	% per month	B3	0	-
56	irf_m	Anbima Market Index of the prefixed federal bonds	index	Anbima	1	% change
57	ima_s	Anbima Market Index of the federal bonds tied to the Selic	index	Anbima	1	% change
58	ima_b	Anbima Market Index of the federal bonds tied to the IPCA	index	Anbima	1	% change
59	ima	General Anbima Market index	index	Anbima	1	% change
60	saving.deposits	Saving deposits – end-of-period balance	R\$ (mil)	BCB	2	% change
61	selic	Selic Basic Interest rate	% per month	BCB	0	-
62	cdi	Cetip DI Interbank Deposits rate	% per month	Cetip	0	-
63	tjlp	TJLP Long-term Interest rate	% per year	BCB	1	-
64	ntnb	3-Year Treasury (real) Rate indexed to the IPCA (NTN-B)	% per year	Anbima	0	-
65	embi	Emerging Markets Bond Index Plus – Brazil	b.p. acc. month	JP Morgan	0	first difference
66	vix	CBOE Volatility Index (VIX)	index	CBOE	0	
H. Credit						
67	cred_total	Credit outstanding – total	R\$ (million)	BCB	2	% change
68	cred_dgp	Credit outstanding as a percentage of GDP	% of GDP	BCB	2	first difference
69	indebt.house1	Household debt to income – total	% of 12m income	BCB	2	first difference
70	indebt.house2	Household debt without mortgage loans	% of 12m income	BCB	2	first difference
I. Government						
71	net_debt_gdp	Net public debt (% GDP) -Consolidated public sector	% of GDP	BCB	2	first difference
72	net_debt	Net public debt – Total – Consolidated public sector	R\$ (million)	BCB	2	% change
73	net_debt.fedgov_bcb	Net public debt – Federal Government and Central Bank	R\$ (million)	BCB	2	% change
74	net_debt.states	Net public debt – State governments	R\$ (million)	BCB	2	% change
75	net_debt.cities	Net public debt – Municipal governments	R\$ (million)	BCB	2	% change
76	primary_result	Primary result – Consolidated public sector	R\$ (million)	BCB	2	% change
77	debt.fedgov_old	Gross general government debt – Method used until 2007	R\$ (million)	BCB	2	% change
78	debt.fedgov_new	Gross general government debt – Method used since 2008	R\$ (million)	BCB	2	% change
79	treasury_emit	National Treasury domestic securities – Total issued	R\$ (million)	BCB	2	% change
80	treasury_mkt	National Treasury domestic securities – Total on market	R\$ (million)	BCB	2	% change
81	treasury_term	National Treasury securities debt – medium term	months	BCB	2	first difference
82	treasury_dur	National Treasury securities debt – medium duration	months	BCB	2	first difference
J. Exchange and International Transactions						
83	reer	Real Effective Exchange Rate	R\$/other	BIS	1	% change
84	usd_brl.end	USD-BRL rate – end-of-period	R\$/US\$	BCB	0	% change
85	usd_brl.av	USD-BRL rate – monthly average	R\$/US\$	BCB	1	% change
86	eur_brl.end	EUR-BRL rate – end-of-period	R\$/€	Bloomberg	0	% change
87	eur_brl.av	EUR-BRL rate – monthly average	R\$/€	Bloomberg	1	% change
88	current.account	Current account – net	US\$ (million)	BCB	2	% change
89	trade.balance	Balance on goods and services – net (Brazilian trade balance)	US\$ (million)	BCB	2	% change
90	exports	Exports	US\$ (million)	BCB	2	% change
91	imports	Imports	US\$ (million)	BCB	2	% change

Notes: “Lag” column indicates the delay for each variable to become available and “Transformations” column indicates transformation implemented to guarantee the stationarity of the series.

C Forecast performance for disaggregates by horizon

C.1 BCB disaggregates before and after the COVID-19 pandemic

Table C1: Out-of-sample RMSE for BCB disaggregate (in terms of RMSE of the AR model): Jan/2014 to Feb/2020, by disaggregate and horizon

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
A. Monitored Prices													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	<i>1.000</i>	1.000	1.000	1.000	1.000
Augmented AR	0.740 ***	0.860 ***	0.971	0.985	1.005	1.010	1.030	1.040	1.040	1.015	1.012	1.024	0.983*
Ridge	<i>0.744</i> ***	0.952	0.990	0.979	0.987	<i>0.975</i>	<i>0.994</i>	<i>0.991</i>	0.998	0.977	0.942 **	0.929 *	0.958 ***
adaLASSO	0.795***	0.929*	<i>0.982</i>	<i>0.985</i>	<i>0.991</i>	0.972	0.999	0.998	1.012	1.010	0.967	0.963	<i>0.970</i> ***
Factor	0.777***	0.937*	1.018	1.011	1.013	0.995	1.019	1.028	1.036	<i>0.994</i>	0.976	<i>0.961</i>	0.983*
FarmPredict	0.815***	0.957	1.008	1.004	1.009	1.011	0.988	0.983	1.013	0.998	0.972	0.967	0.979**
Target Factor	0.796***	<i>0.921</i>	1.004	1.011	1.069	1.076	1.056	1.202	1.151	1.091	1.052	1.012	1.043
CSR	0.950*	0.961*	0.987	0.996	1.013	1.030	1.028	1.041	1.030	1.008	0.997	1.012	1.005
Random Forest	0.913*	0.991	1.032	1.026	1.060	1.076	1.085	1.050	1.047	1.014	<i>0.967</i> *	0.977	1.021
B. Non-Tradables													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	<i>1.000</i>	<i>1.000</i>	1.000	1.000
Augmented AR	<i>0.784</i> ***	0.801**	0.830***	0.795***	0.781***	0.735***	0.819***	0.831**	0.914*	1.063	1.112	1.056	0.867***
Ridge	0.763 ***	<i>0.791</i> ***	<i>0.821</i> ***	<i>0.792</i> ***	<i>0.775</i> ***	0.735***	0.808***	0.820***	<i>0.903</i> **	1.024	1.059	<i>1.015</i>	<i>0.850</i> ***
adaLASSO	0.850**	0.790 **	0.783 ***	0.773 ***	0.746 ***	0.709 ***	<i>0.786</i> ***	<i>0.778</i> ***	0.922*	1.039	1.089	1.087	0.851***
Factor	0.875*	0.837**	0.843**	0.828***	0.791***	0.758***	0.814**	0.813**	0.911**	1.043	1.053	1.026	0.872***
FarmPredict	0.878*	0.827**	0.835***	0.799***	<i>0.770</i> ***	0.741***	0.814***	0.784***	0.923*	1.051	1.070	1.038	0.866***
Target Factor	1.054	0.805**	0.895*	1.027	0.965	0.750***	0.832**	0.907	1.004	1.085	1.058	1.105	0.949**
CSR	0.882***	0.863***	0.887***	0.861***	0.834***	0.799***	0.843***	0.864**	0.979	1.076	1.074	1.035	0.906***
Random Forest	0.820***	0.803***	0.828***	0.797***	0.775***	<i>0.721</i> ***	0.765 ***	0.777 ***	0.841 ***	0.942	0.997	1.038	0.832 ***
C. Tradables													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.786**	<i>0.869</i> ***	0.966	1.047	1.074	1.049	1.000	0.972	0.979	1.011	1.061	1.039	0.996
Ridge	0.763**	0.884**	0.900 **	0.951	<i>0.948</i> *	1.060	0.857**	0.846**	0.869**	0.855 ***	1.047	0.842 ***	0.907***
adaLASSO	<i>0.727</i> ***	0.911**	0.930*	<i>0.948</i> *	0.956*	0.856 **	0.831 ***	0.875**	0.938	<i>0.871</i> **	0.932	<i>0.853</i> ***	0.889***
Factor	0.741***	0.866 ***	0.952	1.002	0.993	0.909	0.858**	<i>0.844</i> **	0.861 **	0.892**	<i>0.885</i> **	0.888**	0.894***
FarmPredict	0.747***	0.894**	0.914**	0.997	0.985	0.910	<i>0.853</i> **	0.843 **	<i>0.868</i> **	0.887***	0.863 **	0.858***	<i>0.887</i> ***
Target Factor	0.722 **	0.936	0.932	1.004	1.119	1.010	0.903*	0.863*	0.892*	1.001	1.000	0.964	0.953**
CSR	0.844***	1.046	1.176	1.065	0.979	0.928	0.965	0.949	0.919*	0.927*	0.971	0.973	0.981
Random Forest	0.768***	0.880**	<i>0.913</i> **	0.923 **	0.888 **	<i>0.880</i> *	0.868**	0.879*	0.905*	0.894**	0.896**	0.883**	0.884 ***

Notes: ***, **, and * indicate that for a specific disaggregate and forecast horizon, a model m performed statistically better than an AR model at 1, 5, and 10% significance levels in a one-tailed Diebold-Mariano test with $\mathbb{H}_0 : \text{MSE}(\hat{\pi}_{i,t+h|t}^m) = \text{MSE}(\pi_{i,t+h|t}^{\text{AR}})$ versus $\mathbb{H}_1 : \text{MSE}(\hat{\pi}_{i,t+h|t}^m) < \text{MSE}(\pi_{i,t+h|t}^{\text{AR}})$. The value highlighted in bold blue indicates the best model for each horizon in terms of RMSE ratio with respect to the AR model, and the values in blue italics indicate the second and third best models.

Table C2: Out-of-sample RMSE for BCB disaggregate (in terms of RMSE of the AR model):
Mar/2020 to Jun/2022, by disaggregate and horizon

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
A. Monitored Prices													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.634 ***	0.955	1.066	1.066	1.049	1.040	1.022	1.029	1.036	1.064	1.034	1.046	1.012
Ridge	0.665 ***	0.955	0.984	0.991	0.981	1.006	1.011	0.978	0.975	0.997	0.977	0.962	0.962 ***
adaLASSO	0.706***	0.999	0.991	0.991	0.981	1.006	1.011	0.978	0.975	0.997	0.977	0.962	0.968 **
Factor	0.679***	1.010	0.997	0.991	0.981	1.006	1.012	0.985	0.971	0.997	0.977	0.962	0.969**
FarmPredict	0.722***	1.007	0.996	0.991	0.981	1.006	1.011	0.980	0.975	0.997	0.977	0.962	0.971**
Target Factor	0.692***	1.160	1.092	1.165	1.056	1.083	1.121	1.083	1.134	1.103	1.101	1.163	1.088
CSR	1.001	1.053	1.063	1.080	1.117	1.081	1.042	1.052	1.106	1.116	1.052	1.047	1.068
Random Forest	0.821**	1.056	1.048	1.101	1.055	1.076	1.115	1.089	1.077	1.058	0.967	0.928	1.036
B. Non-Tradables													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.973	0.889	0.863	0.860	0.855	0.878	0.849	0.907	0.893	0.818**	0.844**	0.891*	0.872***
Ridge	0.957	0.869	0.841	0.833*	0.828	0.858	0.853	0.910	0.894	0.830**	0.848**	0.884**	0.862***
adaLASSO	1.001	0.773**	0.771*	0.784**	0.752**	0.775*	0.862*	1.005	0.989	0.891*	0.805 ***	0.825 ***	0.841***
Factor	0.921	0.750 **	0.754 *	0.763 **	0.720 **	0.803*	0.841 *	0.987	1.031	0.957	0.878**	0.872**	0.851***
FarmPredict	0.938	0.736 **	0.707 **	0.715 **	0.699 **	0.737 **	0.896	1.009	1.018	0.932	0.857***	0.833**	0.832 ***
Target Factor	1.003	0.925	0.898	1.014	0.884	0.883	0.983	1.286	1.070	0.919	0.818*	0.985	0.963
CSR	0.916	0.866**	0.886*	0.928	0.920	0.848*	0.882*	0.903	0.821 **	0.775 ***	0.853***	0.991	0.887***
Random Forest	0.975	0.951	0.799**	0.814**	0.784**	0.743 ***	0.840 **	0.869	0.880	0.810 *	0.780 ***	0.788 ***	0.824 ***
C. Tradables													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.932	0.978	1.033	1.100	1.092	1.053	0.994	0.915*	0.913*	0.955	1.019	1.076	1.012
Ridge	0.912	0.976	1.024	1.020	0.974	1.021	0.953*	0.909***	0.977	1.046	1.006	0.946*	0.983*
adaLASSO	0.842 *	0.989	0.959	0.930**	0.849***	0.830***	0.871***	0.842***	0.891**	0.908*	0.883**	0.810 ***	0.881***
Factor	0.964	1.024	1.000	0.895 ***	0.841***	0.798 ***	0.825***	0.929**	1.018	1.070	1.006	0.928**	0.938***
FarmPredict	0.944	1.079	1.063	0.928**	0.857***	0.825***	0.864***	0.962*	1.027	1.053	1.031	0.949	0.963***
Target Factor	0.938	0.796 **	0.872 *	0.946	0.838**	0.838***	0.694 ***	0.627 ***	0.741 **	0.706 ***	0.975	0.875*	0.820 ***
CSR	0.972	1.044	0.926*	0.918*	0.838 **	0.834***	0.826***	0.792***	0.789***	0.784***	0.858 ***	0.877***	0.863***
Random Forest	0.933	0.941	0.897 **	0.875 ***	0.808 ***	0.753 ***	0.745 ***	0.710 ***	0.734 ***	0.773 ***	0.757 ***	0.703 ***	0.792 ***

Notes: see Table C1.

C.2 Groups

Table C3: Out-of-sample RMSE for IBGE groups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate and horizon

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
A. Foods and beverages (inf .g1)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.959	0.979	1.036	1.034	1.045	1.030	1.006	0.994	0.938	0.929**	1.025	1.069	1.007
Ridge	0.898*	0.916	0.884*	0.903**	0.899**	0.951	0.964	0.965	0.913**	0.831***	0.811***	0.776***	0.886***
adaLASSO	0.805**	0.839***	0.877**	0.877**	0.896**	0.937*	1.012	1.018	0.991	0.896**	0.884*	0.788***	0.900***
Factor	0.841**	0.889**	0.962	0.992	0.964	0.995	1.014	0.966	0.910**	0.833***	0.820***	0.797***	0.911***
FarmPredict	0.847**	0.888*	0.907	0.962	0.933	0.984	1.024	0.966	0.913**	0.872***	0.856**	0.799***	0.909***
Target Factor	0.925	0.908*	0.945	1.010	0.952	0.915*	0.894**	0.924	0.862**	0.966	1.013	0.847***	0.932***
CSR	0.886**	0.955	1.067	1.058	0.983	0.944	0.935*	0.938	1.023	0.886***	0.920	0.856**	0.954**
Random Forest	0.789***	0.831***	0.836***	0.859***	0.848***	0.890**	0.926*	0.927*	0.890**	0.817***	0.792***	0.761***	0.844***
B. Housing (inf .g2)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.780***	0.886**	0.981	1.000	0.992	0.967	1.006	1.012	1.032	0.999	1.017	1.017	0.977**
Ridge	0.873**	0.914**	0.952	0.948**	0.928**	0.883***	0.895***	0.936*	0.900**	0.902**	0.861**	0.914*	0.908***
adaLASSO	0.862***	0.946*	0.952	0.949**	0.948*	0.884***	0.899**	0.935*	0.907**	0.907**	0.860**	0.914*	0.913***
Factor	0.889**	0.967	0.952	0.948**	0.948*	0.884***	0.896**	0.936**	0.897**	0.901**	0.861**	0.913*	0.915***
FarmPredict	0.875**	0.949	0.957	0.979	0.965	0.879***	0.901**	0.944*	0.917*	0.898**	0.862**	0.913*	0.919***
Target Factor	0.807***	0.901**	1.184	1.051	1.044	0.930*	0.981	1.068	0.976	1.033	0.943	0.971	0.992
CSR	0.916***	0.934***	0.986	1.027	0.977	0.943*	0.945**	0.984	0.925**	0.929**	0.907**	0.924*	0.949***
Random Forest	0.898**	0.948	1.001	0.980	0.966	0.944*	0.987	1.002	0.972	0.974	0.903*	0.960	0.961***
C. Household goods (inf .g3)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.053	1.046	1.025	0.984	0.999	1.055	1.064	1.041	1.078	1.091	1.125	1.109	1.062
Ridge	1.140	1.077	1.015	0.976	1.040	0.955	0.926**	0.878***	0.929**	0.908**	0.919**	0.888***	0.956***
adaLASSO	1.034	1.031	1.030	0.996	0.981	0.904**	0.855***	0.800***	0.856***	0.858***	0.854***	0.849***	0.905***
Factor	1.015	0.913*	0.940	0.905**	0.966	0.869***	0.853**	0.885***	0.937*	0.883**	0.885**	0.844***	0.900***
FarmPredict	0.995	0.912*	0.951	0.928*	0.972	0.883*	0.872***	0.890**	0.942*	0.900**	0.906**	0.870***	0.912***
Target Factor	1.030	1.000	0.954	0.914*	1.152	0.905**	0.864***	0.849***	0.879***	0.839***	0.829***	0.800***	0.905***
CSR	1.125	0.893*	0.928*	0.945	0.885**	0.834***	0.800***	0.757***	0.817***	0.807***	0.812***	0.797***	0.849***
Random Forest	1.053	0.914	0.961	0.919*	0.916*	0.839**	0.809***	0.764***	0.800***	0.790***	0.803***	0.777***	0.845***
D. Apparel (inf .g4)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.718***	0.710***	0.731***	0.896**	0.971	0.876**	0.808***	0.801***	0.912*	0.975	1.082	1.026	0.874***
Ridge	0.714***	0.710***	0.709***	0.835**	0.893**	0.834***	0.797***	0.797***	0.858***	0.941*	0.987	0.966	0.836***
adaLASSO	0.752***	0.796***	0.797***	0.798***	0.880***	0.805***	0.827***	0.789***	0.806***	0.974	1.001	0.931*	0.844***
Factor	0.739***	0.816***	0.793***	0.775***	0.878**	0.848***	0.823***	0.849***	0.976	0.929*	0.984	0.924**	0.863***
FarmPredict	0.907**	0.912*	0.867**	0.824***	0.930	0.865**	0.905**	0.908**	1.001	0.950	1.000	0.948	0.920***
Target Factor	0.755***	0.801***	0.800***	0.909*	0.959	0.862*	0.742***	0.775***	0.856***	1.030	1.152	1.014	0.884***
CSR	0.922	0.996	0.938*	0.889**	0.918*	0.861**	0.789***	0.829***	0.879***	0.903*	0.933	0.944*	0.898***
Random Forest	0.831***	0.846***	0.902**	0.897**	0.969	0.869**	0.810***	0.784***	0.823***	0.902**	0.921*	0.906**	0.866***
E. Transportation (inf .g5)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.854***	1.002	1.061	1.017	0.982	0.964	0.977	1.003	0.962	0.994	1.062	1.121	1.001
Ridge	0.859***	1.139	1.116	1.057	0.994	0.970	0.907**	0.868***	0.877***	0.898**	0.997	1.004	0.966**
adaLASSO	0.786***	0.998	1.108	1.070	1.002	0.983	0.927**	0.889***	0.893***	0.917**	0.996	1.001	0.961***
Factor	0.778***	0.966	1.112	1.070	0.997	0.992	0.932**	0.889***	0.891**	0.927**	0.985	0.988	0.959***
FarmPredict	0.790***	1.014	1.097	1.065	0.999	0.989	0.933**	0.910***	0.898**	0.932*	0.986	1.004	0.966***
Target Factor	0.870**	1.126	1.213	1.188	1.037	1.023	0.951	0.902**	0.937	0.943	1.041	1.155	1.026
CSR	0.891**	1.001	0.990	0.994	0.947	0.942**	0.925**	0.876***	0.932*	0.942*	0.995	1.015	0.951***
Random Forest	0.896**	1.035	1.122	1.075	1.012	0.985	0.921**	0.891***	0.909***	0.911**	0.972	0.959	0.968***
F. Medical and personal care (inf .g6)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.991	0.987	0.861***	0.792***	0.781***	0.803***	0.833***	0.872***	0.914**	0.898**	0.900***	0.996	0.881***
Ridge	0.972	0.977**	0.896***	0.822***	0.798***	0.811***	0.846***	0.883***	0.913***	0.921***	0.916***	0.959**	0.889***
adaLASSO	0.974	1.016	0.897**	0.905*	0.931	1.218	1.030	1.051	0.984	1.013	0.944	0.961	0.995
Factor	0.970	0.996	0.977	0.998	0.923*	0.992	0.993	1.008	1.081	1.023	0.981	0.994	0.994
FarmPredict	0.974	1.041	1.045	1.013	0.988	1.023	1.020	1.025	1.043	1.004	0.980	1.023	1.015
Target Factor	0.990	0.958	0.948	0.764***	0.826**	1.026	0.834***	0.926	0.919**	0.877**	0.914*	0.964	0.909***
CSR	1.058	1.050	0.918**	0.857***	1.062	1.062	1.037	1.020	0.957	0.915*	0.919	0.955	0.974*
Random Forest	0.985	1.002	0.899*	0.908*	0.940	1.056	0.954	0.907**	0.929*	0.908**	0.904**	0.969	0.945***

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Table C3: Out-of-sample RMSE for IBGE groups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate and horizon (cont.)

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
G. Personal expenses (inf.g7)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.953	0.962	1.006	0.981	0.997	1.023	1.040	1.039	1.058	1.015	0.977	1.034	1.006
Ridge	0.936	0.884**	0.884***	0.806***	0.878***	0.978	0.967	0.875**	0.949	0.937	0.943	1.021	0.917***
adaLASSO	0.871**	0.840***	0.845***	0.774***	0.840***	0.969	0.949	0.863**	0.927*	0.897**	0.907	1.005	0.886***
Factor	0.864**	0.856***	0.858***	0.783***	0.789***	0.895**	0.905*	0.863**	0.946	0.899**	0.919	0.982	0.876***
FarmPredict	0.900*	0.850***	0.882**	0.781***	0.812***	0.909*	0.941	0.876**	0.931	0.871**	0.874*	1.001	0.882***
Target Factor	0.998	0.973	1.061	0.914	0.971	1.026	0.984	0.931	1.098	0.936	1.070	1.178	1.008
CSR	0.839***	0.946	0.888**	0.822***	0.936	0.957	0.949	0.881**	1.066	0.922	0.936	1.018	0.927***
Random Forest	0.881**	0.851***	0.852***	0.838***	0.943	1.040	1.024	0.903*	0.955	0.933	0.949	1.048	0.931***
H. Education (inf.g8)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.443***	0.396***	0.395***	0.367***	0.351***	0.358***	0.422***	0.411***	0.507***	0.885	0.869	0.879	0.451***
Ridge	0.429***	0.386***	0.377***	0.383***	0.364***	0.380***	0.419***	0.418***	0.495***	0.823*	0.853	0.835	0.446***
adaLASSO	0.432***	0.400***	0.388***	0.385***	0.361***	0.384***	0.420***	0.419***	0.488***	0.874	0.933	0.897	0.457***
Factor	0.432***	0.408***	0.381***	0.374***	0.349***	0.370***	0.409***	0.404***	0.478***	0.878*	0.923*	0.881	0.450***
FarmPredict	0.432***	0.400***	0.387***	0.385***	0.362***	0.384***	0.418***	0.419***	0.486***	0.882	0.930	0.897	0.457***
Target Factor	0.536***	0.425***	0.380***	0.390***	0.377***	0.442***	0.455***	0.457***	0.504***	0.894	0.915	0.943	0.487***
CSR	0.636***	0.823	0.740***	0.663***	0.630***	0.668***	0.654***	0.528***	0.629***	0.924	1.031	1.020	0.698***
Random Forest	0.569***	0.508***	0.538***	0.542***	0.490***	0.543***	0.520***	0.547***	0.607***	0.927	0.915	0.916	0.573***
I. Communication (inf.g9)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.095	1.040	1.002	1.029	1.067	1.060	1.086	1.041	1.077	1.074	1.183	1.030	1.066
Ridge	0.748***	0.806***	0.769***	0.836**	0.822*	0.859**	0.864**	0.740**	0.771***	0.773***	0.842**	0.874*	0.805***
adaLASSO	0.733***	0.810***	0.773***	0.828***	0.823*	0.848**	0.855**	0.732**	0.767***	0.712***	0.819*	0.824*	0.790***
Factor	0.747***	0.820***	0.764***	0.820***	0.830**	0.868**	0.868**	0.733**	0.777***	0.733***	0.787**	0.835*	0.795***
FarmPredict	0.748***	0.832***	0.770***	0.832**	0.818*	0.853**	0.859**	0.735**	0.764***	0.733***	0.817**	0.858*	0.798***
Target Factor	1.031	1.426	1.096	1.210	1.221	1.149	1.552	0.922	1.104	0.916	1.230	1.103	1.164
CSR	0.776***	0.865**	0.842**	0.915*	0.930	0.894**	0.925	0.781***	0.851**	0.889**	0.832*	0.887*	0.863***
Random Forest	0.768***	0.822***	0.831**	0.880*	0.873	0.900*	0.893*	0.762**	0.796**	0.798***	0.921	0.852*	0.838***

Notes: see Table C1.

C.3 Subgroups

Table C4: Out-of-sample RMSE for IBGE subgroups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate and horizon

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
A. Food at home (inf.sg1)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.991	1.105	1.123	1.026	1.053	1.058	1.019	1.064	1.016	0.975	1.152	1.106	1.062
Ridge	0.940	0.832***	0.790***	0.765***	0.766***	0.788***	0.745***	0.730***	0.754***	0.705***	0.719***	0.562***	0.742***
adaLASSO	0.718***	0.831***	0.916	0.790***	0.777***	0.797***	0.759***	0.790***	0.815***	0.759***	0.788***	0.561***	0.766***
Factor	0.799**	0.857***	0.834***	0.797***	0.804***	0.814***	0.777***	0.730***	0.754***	0.705***	0.719***	0.574***	0.751***
FarmPredict	0.815**	0.843***	0.819***	0.792***	0.785***	0.804***	0.776***	0.736***	0.753***	0.758***	0.768***	0.579***	0.757***
Target Factor	0.762**	0.864***	0.948	0.870**	0.829***	0.785***	0.703***	0.718***	0.773***	0.816***	0.831**	0.623***	0.786***
CSR	0.783***	0.878***	0.970	0.885*	0.822***	0.807***	0.730***	0.707***	0.753***	0.714***	0.804***	0.610***	0.779***
Random Forest	0.709***	0.766***	0.761***	0.738***	0.731***	0.747***	0.713***	0.708***	0.744***	0.702***	0.703***	0.554***	0.706***
B. Food away from home (inf.sg2)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.058	0.982	1.106	1.206	1.088	1.051	0.987	1.038	0.938	1.030	1.074	1.097	1.053
Ridge	0.883*	0.899*	0.863**	0.903**	0.869**	0.835***	0.768***	0.769***	0.763**	0.742***	0.672***	0.749***	0.798***
adaLASSO	0.796***	0.803***	0.821***	0.883**	0.795***	0.783***	0.741***	0.766***	0.747***	0.730***	0.628***	0.720***	0.759***
Factor	0.822**	0.822***	0.827***	0.902**	0.893**	0.825***	0.796***	0.738***	0.756***	0.797***	0.714***	0.823***	0.803***
FarmPredict	0.826**	0.846**	0.802***	0.890**	0.868**	0.827***	0.768***	0.760***	0.764**	0.783***	0.712***	0.775***	0.795***
Target Factor	0.881**	0.890**	0.879**	0.901**	0.826***	0.777***	0.787***	0.764***	0.781**	0.791***	0.728***	0.820***	0.811***
CSR	0.813***	0.822***	0.808***	0.839***	0.804***	0.796***	0.754***	0.791***	0.844*	0.750***	0.792**	0.770***	0.796***
Random Forest	0.784***	0.799***	0.788***	0.860***	0.831***	0.811***	0.763***	0.768***	0.766**	0.746***	0.674***	0.754***	0.772***
C. Utilities and maintenance (inf.sg3)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.031	1.061	1.088	1.090	1.007	1.064	1.019	1.074	1.058	1.143	1.157	1.140	1.080
Ridge	0.908	0.853**	0.801**	0.857**	0.797***	0.831***	0.780***	0.814***	0.721***	0.854***	0.805***	0.711***	0.804***
adaLASSO	0.763***	0.788***	0.779***	0.816***	0.798***	0.824***	0.785***	0.811***	0.727***	0.824***	0.798***	0.690***	0.781***
Factor	0.762***	0.806***	0.774***	0.843**	0.785***	0.831***	0.774***	0.819***	0.730***	0.869***	0.825***	0.722***	0.792***
FarmPredict	0.760***	0.809***	0.780***	0.839***	0.785***	0.820***	0.783***	0.820***	0.729***	0.868***	0.831**	0.730***	0.793***
Target Factor	0.923	0.766***	0.763***	0.958	0.848***	1.026	0.841***	0.832***	0.752***	0.884**	0.844**	0.863**	0.857***
CSR	0.836	0.849***	0.781***	0.781***	0.747***	0.784***	0.776***	0.793***	0.712***	0.831***	0.822**	0.754***	0.785***
Random Forest	0.805**	0.770***	0.711***	0.771***	0.772***	0.806***	0.783***	0.825***	0.726***	0.847***	0.792***	0.675***	0.770***
D. Domestic fuels and energy (inf.sg4)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.802***	0.912**	1.033	1.050	0.976	0.965*	1.003	1.025	1.074	1.035	1.056	1.055	1.000
Ridge	0.776***	0.760***	0.849***	0.838***	0.832***	0.780***	0.763***	0.837***	0.812***	0.772***	0.723***	0.815***	0.794***
adaLASSO	0.756***	0.782***	0.887**	0.838***	0.856***	0.780***	0.763***	0.843***	0.816***	0.769***	0.743***	0.832***	0.803***
Factor	0.773***	0.786***	0.887**	0.836***	0.857***	0.780***	0.763***	0.843***	0.817***	0.773***	0.743***	0.834***	0.805***
FarmPredict	0.771***	0.784***	0.874**	0.848***	0.889**	0.788***	0.784***	0.840***	0.816***	0.775***	0.753***	0.834***	0.810***
Target Factor	0.720***	0.772***	1.545	0.984	0.907**	0.898**	0.901**	0.964	0.985	0.931	0.833**	0.911*	0.954
CSR	0.788***	0.831***	0.905**	0.892**	0.894**	0.860**	0.835**	0.930*	0.849***	0.815***	0.780***	0.847***	0.850***
Random Forest	0.789***	0.816***	0.925*	0.878***	0.895**	0.866***	0.875***	0.920**	0.888**	0.841**	0.769***	0.866**	0.858***
E. Furniture and fixtures (inf.sg5)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.071	1.015	1.049	1.129	1.013	1.003	1.025	1.056	1.039	1.138	1.159	1.081	1.063
Ridge	1.152	0.976	0.888**	0.867**	0.832***	0.734***	0.775***	0.842***	0.795***	0.894**	0.871***	0.816***	0.854***
adaLASSO	0.888*	0.910*	0.888**	0.853***	0.797***	0.724***	0.770***	0.851***	0.784***	0.869***	0.837***	0.799***	0.823***
Factor	0.878*	0.876**	0.832***	0.905**	0.868***	0.737***	0.780***	0.856***	0.812***	0.896**	0.862***	0.823***	0.838***
FarmPredict	0.882*	0.895***	0.819***	0.882**	0.844***	0.734***	0.784***	0.865***	0.821***	0.914**	0.877***	0.825***	0.839***
Target Factor	0.927	0.927	0.844**	0.915*	0.989	0.816***	0.813***	0.822***	0.760***	0.858***	0.835***	0.847**	0.858***
CSR	0.899*	0.848***	0.824***	0.826***	0.809***	0.732***	0.767***	0.815***	0.758***	0.830***	0.811***	0.782***	0.801***
Random Forest	0.926	0.905**	0.917*	0.968	0.964	0.820**	0.843**	0.861***	0.767***	0.791***	0.753***	0.712***	0.849***
F. Appliances (inf.sg6)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.052	1.099	1.007	1.004	1.031	1.084	1.111	1.073	1.153	1.106	1.083	1.165	1.085
Ridge	0.954	0.875**	0.850***	0.883**	0.883**	0.824***	0.775***	0.735***	0.780***	0.803***	0.795***	0.839**	0.827***
adaLASSO	0.911	0.865**	0.836***	0.876**	0.880**	0.814***	0.757***	0.714***	0.789**	0.831**	0.829**	0.868**	0.826***
Factor	0.903*	0.817***	0.840***	0.850***	0.876**	0.794***	0.768***	0.732***	0.771***	0.804***	0.798***	0.830**	0.810***
FarmPredict	0.903*	0.820***	0.858**	0.846***	0.879**	0.811***	0.787***	0.728***	0.780***	0.800***	0.797***	0.831**	0.815***
Target Factor	0.874*	0.870**	0.848***	0.897**	0.932	0.800***	0.861**	0.786***	0.749***	0.781***	0.824***	0.852**	0.836***
CSR	1.057	0.836**	0.815***	0.886**	0.852***	0.798***	0.750***	0.786***	0.753***	0.772***	0.766***	0.833**	0.819***
Random Forest	0.827**	0.794***	0.781***	0.806***	0.800***	0.755***	0.718***	0.688***	0.712***	0.731***	0.734***	0.795***	0.757***

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Table C4: Out-of-sample RMSE for IBGE subgroups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate and horizon (cont.)

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
G. Household operations (inf.sg7)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.079	1.047	0.995	1.062	0.989	1.064	1.066	1.103	1.107	1.040	0.995	1.108	1.050
Ridge	0.705***	0.700***	0.607***	0.681***	0.649***	0.733***	0.817***	0.730***	0.738***	0.630***	0.594***	0.685***	0.681***
adaLASSO	0.720***	0.708***	0.606***	0.679***	0.653***	0.749***	0.798***	0.714***	0.716***	0.632***	0.592***	0.651***	0.677***
Factor	0.712***	0.711***	0.623***	0.696***	0.646***	0.733***	0.820***	0.731***	0.724***	0.632***	0.594***	0.682***	0.684***
FarmPredict	0.709***	0.721***	0.613***	0.688***	0.652***	0.735***	0.817***	0.725***	0.738***	0.640***	0.607***	0.695***	0.687***
Target Factor	0.839**	0.772***	0.698***	0.858*	0.783***	0.837***	0.926	0.822**	0.851**	0.811**	0.688**	0.938	0.812**
CSR	0.726***	0.711***	0.657***	0.712***	0.670***	0.697***	0.766***	0.741***	0.733***	0.612***	0.579***	0.671***	0.683***
Random Forest	0.668***	0.684***	0.593***	0.657***	0.610***	0.684***	0.756***	0.695***	0.709***	0.607***	0.556***	0.663***	0.650***
H. Clothes (inf.sg8)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.824***	0.874*	0.804***	0.838***	0.891*	0.871*	0.883*	0.897*	0.924	1.024	1.014	0.940	0.902***
Ridge	0.804***	0.887**	0.781***	0.861**	0.857*	0.966	0.902*	0.875**	0.880**	0.939	0.938	0.912	0.886***
adaLASSO	0.745***	0.818***	0.704***	0.629***	0.684***	0.701***	0.770***	0.760***	0.687***	0.775**	0.796***	0.781***	0.737***
Factor	0.722***	0.793***	0.720***	0.631***	0.694***	0.711***	0.754***	0.845***	0.847**	0.779**	0.806***	0.772***	0.759***
FarmPredict	0.873**	0.896*	0.789***	0.688***	0.735***	0.721***	0.819***	0.876**	0.890*	0.781**	0.834***	0.800***	0.809***
Target Factor	0.723***	0.715***	0.659***	0.646***	0.704***	0.754***	0.701***	0.686***	0.779***	0.769***	0.829**	0.782***	0.731***
CSR	0.966	1.039	0.784***	0.745***	0.791**	0.780***	0.818***	0.802***	0.791***	0.755**	0.772***	0.773***	0.810***
Random Forest	0.813***	0.834***	0.776***	0.698***	0.745***	0.731***	0.780***	0.765***	0.725***	0.746**	0.762***	0.749***	0.759***
I. Footwear and accessories (inf.sg9)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.023	1.063	0.957	0.959	1.011	1.049	0.978	0.932*	1.000	1.145	1.218	1.062	1.040
Ridge	0.938	0.925	0.874*	0.776**	0.884	0.951	0.838**	0.794***	0.800**	0.730*	0.791*	0.798***	0.832***
adaLASSO	0.812***	0.782***	0.780***	0.672***	0.769*	0.860**	0.876**	0.789***	0.794**	0.668*	0.734**	0.738***	0.768***
Factor	0.830***	0.860**	0.792***	0.670***	0.786*	0.914*	0.862**	0.787***	0.817*	0.670*	0.717**	0.731***	0.780***
FarmPredict	0.884**	0.919	0.837**	0.676***	0.773*	0.885*	0.805***	0.792***	0.797**	0.676*	0.727**	0.739***	0.784***
Target Factor	0.741***	0.788***	0.783***	0.741***	0.818*	0.911*	0.831**	0.810**	0.841*	0.851	0.731**	0.765***	0.802***
CSR	0.889	1.004	0.802***	0.685***	0.789*	0.910*	0.831***	0.824**	0.719**	0.670*	0.737**	0.736***	0.791***
Random Forest	0.769***	0.779***	0.792***	0.681***	0.763*	0.840**	0.752***	0.715***	0.732**	0.652**	0.698**	0.707***	0.734***
J. Jewelry (inf.sg10)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.084	1.097	1.129	1.074	1.119	1.029	1.106	1.110	0.940	1.166	1.137	1.280	1.106
Ridge	0.714	0.691***	0.704***	0.697***	0.681***	0.677***	0.662***	0.717***	0.678***	0.697***	0.666***	0.739***	0.693***
adaLASSO	0.678***	0.652***	0.716	0.732***	0.713***	0.727***	0.693***	0.762***	0.722***	0.739***	0.691***	0.827***	0.721***
Factor	0.722**	0.705***	0.732***	0.751***	0.764***	0.723***	0.709***	0.745***	0.683***	0.697***	0.671***	0.810***	0.725***
FarmPredict	0.723**	0.704***	0.731***	0.740***	0.725***	0.733***	0.726***	0.822***	0.679***	0.705***	0.680***	0.774***	0.728***
Target Factor	0.771**	0.777***	0.811	0.798**	0.765***	0.802***	0.867***	0.883***	0.807***	0.788***	0.801**	0.814***	0.808***
CSR	0.705***	0.683**	0.710	0.699*	0.717***	0.712***	0.701***	0.776***	0.761***	0.746***	0.726***	0.822***	0.730***
Random Forest	0.683***	0.664***	0.674***	0.682***	0.679***	0.687***	0.682***	0.746***	0.706***	0.722***	0.675***	0.730***	0.694***
K. Fabrics (inf.sg11)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.170	1.134	1.125	1.114	1.155	1.140	1.130	1.099	1.076	1.343	1.136	1.041	1.147
Ridge	0.811	0.833***	0.759***	0.766***	0.724***	0.734***	0.820***	0.807***	0.789***	0.643***	0.595***	0.690***	0.736***
adaLASSO	0.799***	0.833***	0.758	0.726***	0.730***	0.698***	0.766***	0.770***	0.755***	0.600***	0.547***	0.639***	0.706***
Factor	0.773**	0.831***	0.747***	0.759***	0.727***	0.738***	0.822***	0.816***	0.804***	0.648***	0.611***	0.685***	0.736***
FarmPredict	0.790**	0.818***	0.747***	0.761***	0.724***	0.743***	0.853***	0.836***	0.794***	0.643***	0.605***	0.693***	0.739***
Target Factor	0.825**	0.900***	0.796	0.750**	0.723***	0.762***	0.828***	0.972***	0.811***	0.719***	0.600***	0.740***	0.774***
CSR	0.820***	0.838**	0.757	0.846*	0.736***	0.719***	0.769***	0.756***	0.706***	0.584***	0.592***	0.677***	0.722***
Random Forest	0.729***	0.745***	0.694***	0.695***	0.660***	0.678***	0.746***	0.733***	0.727***	0.604***	0.550***	0.638***	0.673***
L. Transportation (inf.sg12)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.858	1.000	1.116	1.063	1.022	1.000	0.977	1.042	1.052	1.039	1.083	1.135	1.037
Ridge	0.977**	0.915*	0.963**	0.925**	0.885**	0.880***	0.832***	0.830***	0.839**	0.787***	0.877***	0.844***	0.874***
adaLASSO	0.709***	0.830***	0.955***	0.955**	0.907***	0.880***	0.825***	0.830***	0.852***	0.804***	0.900***	0.878***	0.861***
Factor	0.683**	0.821***	0.967***	0.946**	0.890**	0.900***	0.837***	0.839***	0.853***	0.818***	0.908***	0.866***	0.863***
FarmPredict	0.705**	0.827**	0.951***	0.930**	0.887**	0.895***	0.836***	0.842***	0.858**	0.817***	0.899***	0.871***	0.861***
Target Factor	0.783**	0.912*	1.071**	1.082**	0.938***	0.966***	0.900***	0.855***	0.877**	0.774***	0.930***	0.999***	0.923***
CSR	0.812***	0.890***	0.903***	0.901***	0.881***	0.901***	0.859***	0.843***	0.830*	0.847***	0.884**	0.859***	0.867***
Random Forest	0.789***	0.861***	0.954***	0.945***	0.897***	0.907***	0.844***	0.852***	0.864**	0.780***	0.840***	0.806***	0.860***
M. Pharmaceutical and optical products (inf.sg13)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.086	1.047	0.812	0.797	0.800	0.814	0.852	0.868	0.910	0.885	0.907	1.042	0.889
Ridge	1.273	1.217**	0.937**	0.895**	0.903***	0.897***	0.847***	0.882***	0.990***	0.902***	0.853***	1.010***	0.941***
adaLASSO	0.941***	0.981***	0.814***	0.822***	0.925***	0.909***	0.874***	0.755***	0.755***	0.796***	0.718***	0.774***	0.827***
Factor	0.965***	0.996***	1.020***	0.932**	0.943***	0.918***	0.871***	0.811***	0.947***	0.822***	0.798***	0.780***	0.892***
FarmPredict	0.971***	0.984***	1.019***	0.924***	0.946***	0.914***	0.874***	0.767***	0.865***	0.805**	0.795**	0.780***	0.877***
Target Factor	0.951	0.937***	0.740***	0.698	0.717***	0.671	0.669***	0.681***	0.697***	0.727**	0.714**	0.775**	0.729***
CSR	1.126	1.061***	0.805***	0.922***	0.943***	0.928***	0.875***	0.764***	0.755***	0.734***	0.708**	0.788***	0.848***
Random Forest	1.043**	0.967***	0.743***	0.809***	0.955***	0.900***	0.802***	0.697***	0.737***	0.694***	0.673***	0.756***	0.798***

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Table C4: Out-of-sample RMSE for IBGE subgroups (in terms of RMSE of the AR model): Jan/2014 to Jun/2022, by disaggregate and horizon (cont.)

Estimator/Model	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	all h
N. Medical services (inf. sg14)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.009***	1.012**	1.018	0.989	1.013	1.035*	1.002	1.025	1.007	1.028	0.999	0.980	1.010
Ridge	0.983***	0.984***	0.989***	0.987***	0.979***	0.971***	1.012***	0.984***	1.003***	1.006***	0.978***	0.977***	0.987***
adaLASSO	0.952***	0.903***	0.904**	1.009***	1.025***	0.978***	1.039***	0.964***	0.991***	1.032***	0.964***	0.998***	0.979***
Factor	0.945***	0.901***	0.934**	1.011***	0.982***	0.980***	1.008***	0.962***	0.982***	1.012***	0.957***	0.979***	0.970***
FarmPredict	0.931***	0.906***	0.928**	1.043***	0.988**	0.995***	1.046***	0.967***	0.972***	1.035***	0.948***	0.980***	0.978***
Target Factor	0.949***	1.019***	0.967	0.932	0.979**	0.919*	0.989**	0.852	0.938	0.984	0.945**	1.009*	0.958
CSR	0.938***	0.899***	0.902**	0.979**	0.960**	0.966**	1.020***	0.973*	0.976***	1.030***	0.987***	0.998***	0.967***
Random Forest	0.947***	0.944***	0.916*	0.967***	0.957**	0.967***	1.015***	0.988**	0.975**	0.975**	0.974***	1.023**	0.969***
O. Personal care (inf. sg15)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.973	0.995	1.034	1.059	1.042	1.019	1.001	0.984	1.071	1.054	1.005	1.021	1.023
Ridge	0.905	0.901	0.880**	0.863**	0.858***	0.953***	0.982***	0.923***	0.901***	0.907**	0.914***	0.833***	0.899***
adaLASSO	0.922*	0.893*	0.895**	0.881***	0.870***	0.954***	0.975***	0.918***	0.878***	0.889***	0.892***	0.833***	0.898***
Factor	0.881*	0.905**	0.882***	0.867**	0.871***	0.950***	0.981***	0.917***	0.924***	0.896**	0.910***	0.834***	0.899***
FarmPredict	0.885*	0.900**	0.869***	0.851**	0.849***	0.950***	0.983***	0.925***	0.916***	0.906**	0.918***	0.832***	0.896***
Target Factor	0.950	0.881	0.921**	0.883*	0.881	0.898***	0.923***	0.852***	0.935***	0.966***	1.084***	0.903**	0.924***
CSR	0.924*	0.903***	0.888***	0.878***	0.889***	0.952***	0.966***	0.898***	0.863***	0.869***	0.915***	0.840***	0.897***
Random Forest	0.899	0.925**	0.924*	0.903	0.868	0.951***	0.988**	0.922***	0.913***	0.928***	0.926***	0.831***	0.913***
P. Personal services (inf. sg16)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.097	1.138	1.027	1.068	1.184	1.138	1.066	0.987	1.225	1.260	1.098	1.151	1.117
Ridge	0.864	0.778**	0.765***	0.786**	0.826**	0.733***	0.781***	0.727***	0.821***	0.972***	0.817***	0.675**	0.786***
adaLASSO	0.698	0.698**	0.707***	0.767**	0.816**	0.712***	0.753***	0.689***	0.663**	0.769***	0.649**	0.663**	0.713***
Factor	0.711*	0.719***	0.734***	0.755***	0.833**	0.725***	0.749***	0.700***	0.651***	0.778***	0.657***	0.649**	0.719***
FarmPredict	0.742*	0.758***	0.747**	0.793***	0.821**	0.707***	0.757***	0.711***	0.656***	0.783***	0.666***	0.660**	0.730***
Target Factor	0.792*	0.742**	0.807***	0.878**	0.786	0.786***	0.977***	0.735***	0.781***	0.838***	0.742***	0.804**	0.807***
CSR	0.655	0.571**	0.609***	0.636**	0.642***	0.605***	0.630***	0.629***	0.723***	0.929***	0.728***	0.646**	0.662***
Random Forest	0.606**	0.564***	0.576***	0.619***	0.659***	0.559***	0.629***	0.605***	0.716***	0.884***	0.731***	0.619***	0.643***
Q. Recreation and tobacco (inf. sg17)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.988	1.019	1.021	1.135	1.042	1.032	1.017	1.076	1.144	1.129	1.029	1.056	1.061
Ridge	0.730***	0.786***	0.791***	0.740***	0.695***	0.773***	0.749***	0.756***	0.703***	0.684***	0.774***	0.747***	0.742***
adaLASSO	0.718***	0.791***	0.798***	0.744***	0.676***	0.759***	0.740***	0.752***	0.699***	0.678***	0.772***	0.750***	0.737***
Factor	0.705***	0.781***	0.776***	0.742***	0.654***	0.756***	0.737***	0.756***	0.705***	0.690***	0.792***	0.744***	0.734***
FarmPredict	0.719***	0.783***	0.787***	0.740***	0.666***	0.781***	0.767***	0.753***	0.698***	0.680***	0.772***	0.747***	0.738***
Target Factor	0.781**	0.810***	0.896***	1.108**	0.809***	0.824***	0.839	0.811**	0.754**	0.699**	0.835**	0.773	0.831***
CSR	0.723***	0.854***	0.795***	0.732***	0.727***	0.774***	0.767***	0.862***	0.794***	0.729***	0.796***	0.778***	0.776***
Random Forest	0.753***	0.796***	0.784***	0.776***	0.757***	0.850***	0.815***	0.769***	0.693***	0.652***	0.763***	0.747***	0.761***
R. Courses, reading, and stationery (inf. sg18)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	0.419***	0.411*	0.387***	0.330***	0.336*	0.328*	0.397*	0.443*	0.543	0.830	0.869	0.928	0.436***
Ridge	0.410***	0.364**	0.345***	0.330**	0.349*	0.342	0.389*	0.440**	0.484**	0.765	0.794	0.844	0.415***
adaLASSO	0.346***	0.350***	0.328***	0.303***	0.307***	0.311***	0.354***	0.400***	0.453***	0.811**	0.823***	0.819***	0.389***
Factor	0.353***	0.358***	0.338***	0.295***	0.304***	0.301***	0.345***	0.372***	0.416***	0.796**	0.810***	0.820***	0.383***
FarmPredict	0.346**	0.350**	0.328***	0.303***	0.308***	0.311***	0.354***	0.400***	0.452*	0.803**	0.821***	0.822***	0.389***
Target Factor	0.384***	0.369***	0.335***	0.295***	0.330***	0.304***	0.395***	0.426***	0.456***	0.787**	0.845**	0.761***	0.400***
CSR	0.466	0.801	0.603***	0.508**	0.550**	0.528***	0.568***	0.502***	0.693***	0.834**	0.917***	0.910***	0.604***
Random Forest	0.454***	0.455***	0.445***	0.425***	0.414***	0.420***	0.428***	0.497***	0.556***	0.848**	0.802***	0.843***	0.480***
S. Communication (inf. sg19)													
AR	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Augmented AR	1.141	1.108	1.062	1.064	1.058	1.091	0.972	0.961*	0.919	0.928	0.988	1.030	1.022
Ridge	0.653	0.705	0.618*	0.642**	0.622	0.664	0.677**	0.616***	0.553**	0.574*	0.535*	0.573***	0.613***
adaLASSO	0.638***	0.700***	0.613***	0.636***	0.616*	0.656**	0.673**	0.609***	0.551**	0.528*	0.506**	0.537***	0.599***
Factor	0.650***	0.702**	0.619***	0.647***	0.621*	0.657*	0.674**	0.609***	0.569*	0.539*	0.502**	0.545***	0.605***
FarmPredict	0.645**	0.720	0.621**	0.643***	0.622*	0.660*	0.673***	0.611***	0.549**	0.544*	0.514**	0.532***	0.604***
Target Factor	0.856***	1.154***	0.874***	0.986***	0.912*	0.864*	0.934**	0.732**	0.667*	0.695	0.708**	0.660***	0.829***
CSR	0.672	0.761	0.646***	0.684***	0.680**	0.679*	0.723***	0.631**	0.599**	0.657*	0.511**	0.663***	0.653***
Random Forest	0.656***	0.711***	0.652***	0.662***	0.668*	0.694**	0.734***	0.649***	0.580**	0.602**	0.576**	0.560***	0.639***

Notes: see Table C1.

D Frequency of models with least squared forecast error

Figure D1: Frequency each model attained the least forecast squared error for BCB disaggregation, by horizon and disaggregate (%)

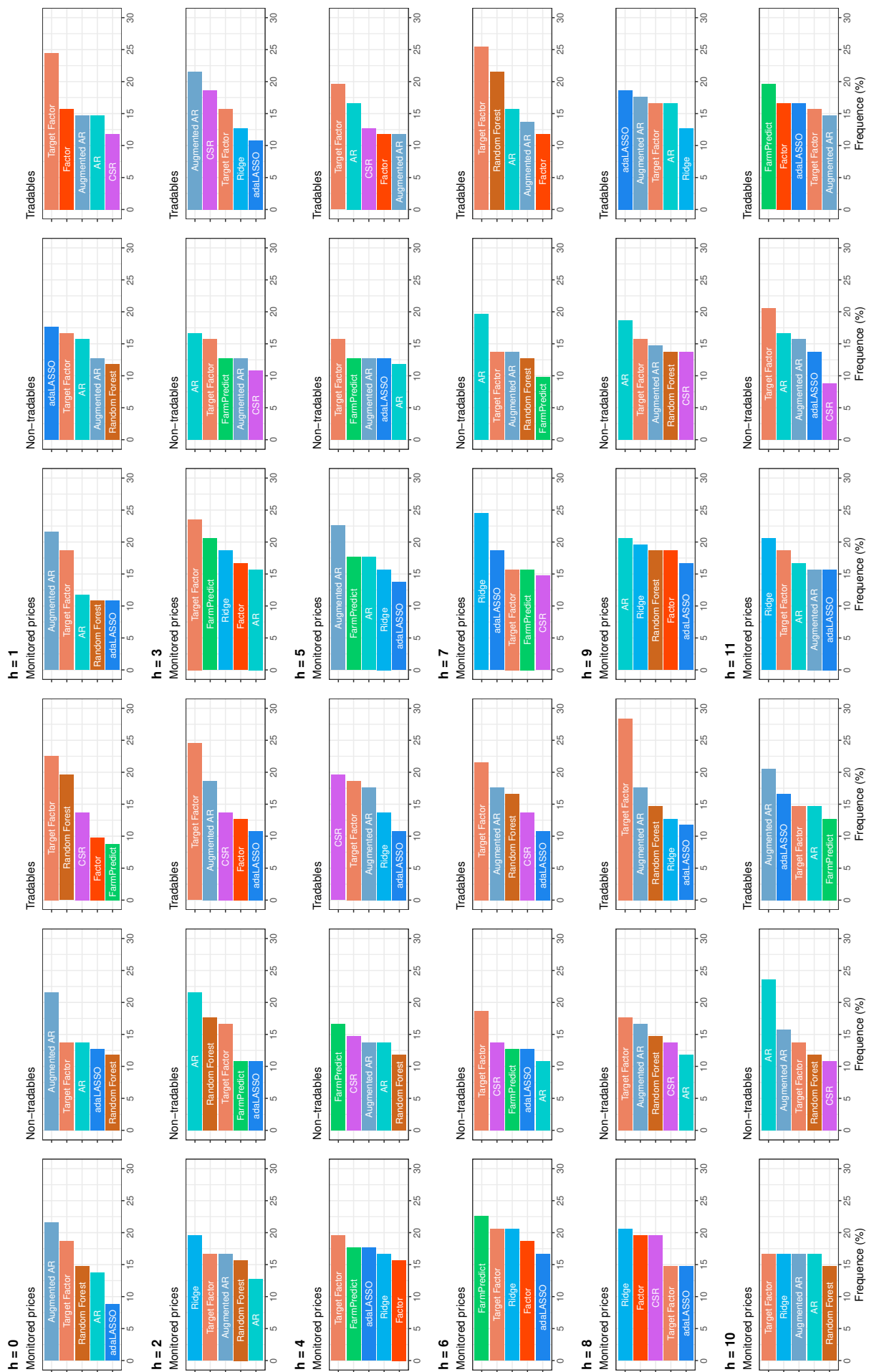


Figure D2: Frequency each model attained the least forecast squared error for BCB disaggregation, by disaggregate and stacking the horizons (%)

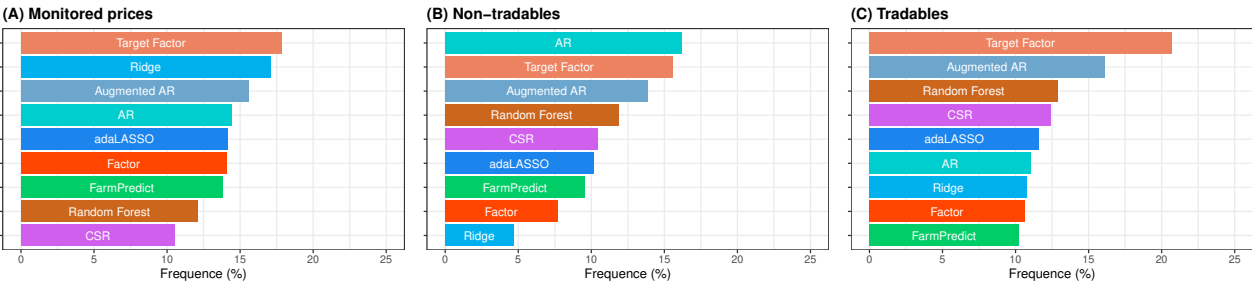


Figure D3: Frequency each model attained the least forecast squared error for IBGE groups, by disaggregate and stacking the horizons (%)

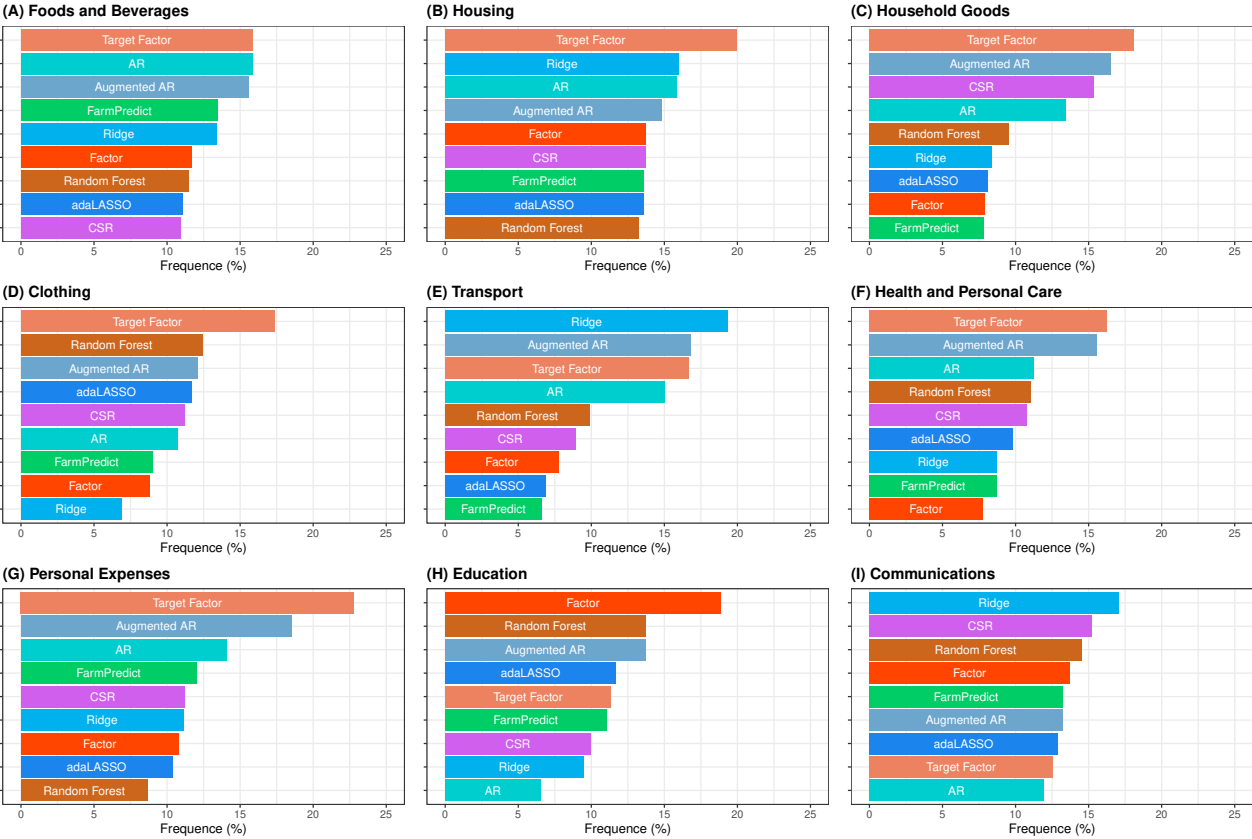
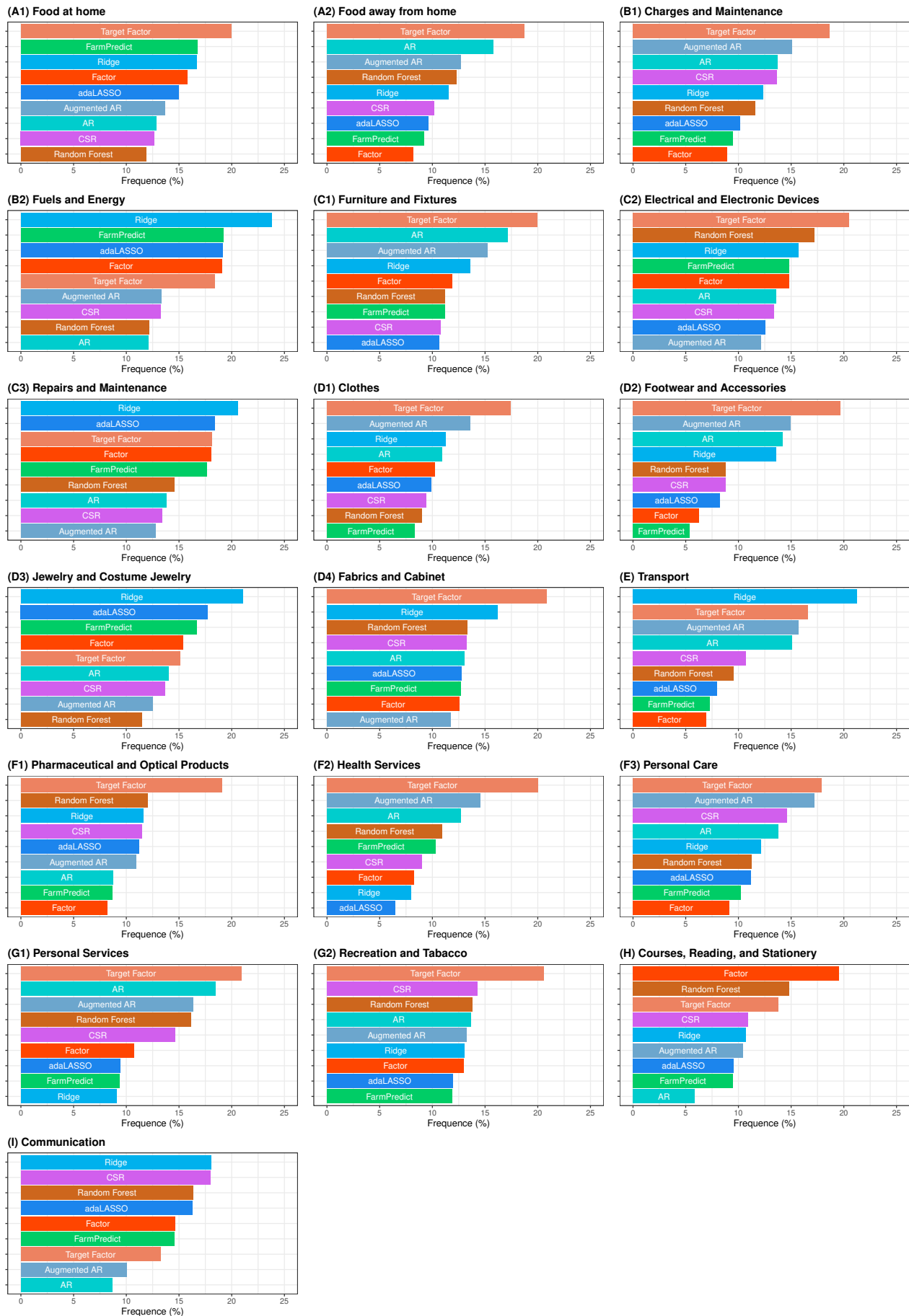


Figure D4: Frequency each model attained the least forecast squared error for IBGE subgroups, by disaggregate and stacking the horizons (%)



References

- ALTUG, S., AND C. ÇAKMAKLI (2016): "Forecasting inflation using survey expectations and target inflation: Evidence for Brazil and Turkey," *International Journal of Forecasting*, 32(1), 138–153.
- ANG, A., G. BEKAERT, AND M. WEI (2007): "Do macro variables, asset markets, or surveys forecast inflation better?," *Journal of Monetary Economics*, 54(4), 1163–1212.
- ARAUJO, G. S., AND W. P. GAGLIANONE (2023): "Machine learning methods for inflation forecasting in Brazil: New contenders versus classical models," *Latin American Journal of Central Banking*, 4(2), 1–29.
- ARON, J., AND J. MUELLBAUER (2012): "Improving forecasting in an emerging economy, South Africa: Changing trends, long run restrictions and disaggregation," *International Journal of Forecasting*, 28(2), 456–476.
- BAI, J., AND S. NG (2002): "Determining the number of factors in approximate factor models," *Econometrica*, 70(1), 191–221.
- (2008): "Forecasting economic time series using targeted predictors," *Journal of Econometrics*, 146(2), 304–317.
- (2009): "Boosting diffusion indices," *Journal of Applied Econometrics*, 24(4), 607–629.
- BAÑBURA, M., D. LEIVA-LEÓN, AND J.-O. MENZ (2021): "Do inflation expectations improve model-based inflation forecasts?," ECB Working Paper n. 2604.
- BAŞTÜRK, N., C. ÇAKMAKLI, S. P. CEYHAN, AND H. K. VAN DIJK (2014): "Posterior-predictive evidence on US inflation using extended new Keynesian Phillips curve models with non-filtered data," *Journal of Applied Econometrics*, 29(7), 1164–1182.
- BAYBUZA, I. (2018): "Inflation forecasting using machine learning methods," *Russian Journal of Money and Finance*, 77(4), 42–59.
- BENALAL, N., J. L. DIAZ DEL HOYO, B. LANDAU, M. ROMA, AND F. SKUDELNY (2004): "To aggregate or not to aggregate? Euro area inflation forecasting," ECB Working Paper n. 374.
- BERMINGHAM, C., AND A. D'AGOSTINO (2014): "Understanding and forecasting aggregate and disaggregate price dynamics," *Empirical Economics*, 46(2), 765–788.
- BLANCHARD, O., AND J. GALÍ (2007): "Real wage rigidities and the New Keynesian model," *Journal of Money, Credit and Banking*, 39, 35–65.
- BREIMAN, L. (2001): "Random forests," *Machine learning*, 45(1), 5–32.
- BRUNEAU, C., O. DE BANDT, A. FLAGEOLLET, AND E. MICHAUX (2007): "Forecasting inflation using economic indicators: the case of France," *Journal of Forecasting*, 26(1), 1–22.
- CAPISTRÁN, C., C. CONSTANDSE, AND M. RAMOS-FRANCIA (2010): "Multi-horizon inflation forecasts using disaggregated data," *Economic Modelling*, 27(3), 666–677.
- CARLO, T. C., AND E. F. MARÇAL (2016): "Forecasting Brazilian inflation by its aggregate and disaggregated data: a test of predictive power by forecast horizon," *Applied Economics*, 48(50), 4846–4860.
- CHOUDHARY, M. A., AND A. HAIDER (2012): "Neural Network Models for Inflation Forecasting: An Appraisal," *Applied Economics*, 44(20), 2631–2635.
- CROUSHORE, D. (2010): "An evaluation of inflation forecasts from surveys using real-time data," *The BE Journal of Macroeconomics*, 10(1).
- DEL NEGRO, M., AND S. EUSEPI (2011): "Fitting observed inflation expectations," *Journal of Economic Dynamics and Control*, 35(12), 2105–2131.
- DIEBOLD, F. X., AND R. S. MARIANO (1995): "Comparing Predictive Accuracy," *Journal of Business & Economic Statistics*, 13(3).
- DUARTE, C., AND A. RUA (2007): "Forecasting inflation through a bottom-up approach: How bottom is bottom?," *Economic Modelling*, 24(6), 941–953.

- ELLIOTT, G., A. GARGANO, AND A. TIMMERMANN (2013): "Complete subset regressions," *Journal of Econometrics*, 177(2), 357–373.
- (2015): "Complete subset regressions with large-dimensional sets of predictors," *Journal of Economic Dynamics and Control*, 54, 86–110.
- ESPASA, A., AND R. ALBACETE (2007): "Econometric modelling for short-term inflation forecasting in the euro area," *Journal of Forecasting*, 26(5), 303–316.
- ESPASA, A., E. SENRA, AND R. ALBACETE (2002): "Forecasting inflation in the European Monetary Union: A disaggregated approach by countries and by sectors," *The European Journal of Finance*, 8(4), 402–421.
- FAN, J., R. MASINI, AND M. C. MEDEIROS (2021): "Bridging factor and sparse models," Available at SSRN 3789141.
- FAUST, J., AND J. H. WRIGHT (2013): "Forecasting Inflation," in *Handbook of Economic Forecasting*, ed. by G. Elliott, and A. Timmermann, vol. 2A, chap. 1, pp. 3–56. Elsevier, North Holland.
- FLORIDO, G. P. (2021): "Essays on Inflation Forecasting," Ph.D. thesis, Sao Paulo School of Economics, FGV.
- FORNI, M., M. HALLIN, M. LIPPI, AND L. REICHLIN (2003): "Do financial variables help forecasting inflation and real activity in the euro area?," *Journal of Monetary Economics*, 50(6), 1243–1255.
- FULTON, C., AND K. HUBRICH (2021): "Forecasting US inflation in real time," *Econometrics*, 9(36), 1–20.
- GALÍ, J., AND M. GERTLER (1999): "Inflation dynamics: A structural econometric analysis," *Journal of Monetary Economics*, 44(2), 195–222.
- GARCIA, M. G., M. C. MEDEIROS, AND G. F. VASCONCELOS (2017): "Real-time inflation forecasting with high-dimensional models: The case of Brazil," *International Journal of Forecasting*, 33(3), 679–693.
- HENDRY, D. F., AND K. HUBRICH (2011): "Combining disaggregate forecasts or combining disaggregate information to forecast an aggregate," *Journal of Business & Economic Statistics*, 29(2), 216–227.
- HUBRICH, K. (2005): "Forecasting euro area inflation: Does aggregating forecasts by HICP component improve forecast accuracy?," *International Journal of Forecasting*, 21(1), 119–136.
- IBARRA, R. (2012): "Do disaggregated CPI data improve the accuracy of inflation forecasts?," *Economic Modelling*, 29(4), 1305–1313.
- INOUE, A., AND L. KILIAN (2008): "How Useful Is Bagging in Forecasting Economic Time Series? A Case Study of U.S. Consumer Price Inflation," *Journal of the American Statistical Association*, 103(482), 511–522.
- LÜTKEPOHL, H. (1984): "Linear transformations of vector ARMA processes," *Journal of Econometrics*, 26(3), 283–293.
- MEDEIROS, M. C., G. VASCONCELOS, AND E. FREITAS (2016): "Forecasting Brazilian inflation with high-dimensional models," *Brazilian Review of Econometrics*, 36(2), 223–254.
- MEDEIROS, M. C., G. F. VASCONCELOS, Á. VEIGA, AND E. ZILBERMAN (2021): "Forecasting inflation in a data-rich environment: the benefits of machine learning methods," *Journal of Business & Economic Statistics*, 39(1), 98–119.
- MOSER, G., F. RUMLER, AND J. SCHARLER (2007): "Forecasting Austrian Inflation," *Economic Modelling*, 24(3), 470–480.
- MOSHIRI, S., AND N. CAMERON (2000): "Neural Network versus Econometric Models in Forecasting Inflation," *Journal of Forecasting*, 19(3), 201–217.
- NAKAMURA, E. (2005): "Inflation forecasting using a neural network," *Economics Letters*, 86(3), 373–378.
- STOCK, J. H., AND M. W. WATSON (1999): "Forecasting inflation," *Journal of Monetary Economics*, 44(2), 293–335.
- (2002): "Macroeconomic forecasting using diffusion indexes," *Journal of Business & Economic Statistics*, 20(2), 147–162.

- THOMAS, L. B. (1999): "Survey measures of expected US inflation," *Journal of Economic Perspectives*, 13(4), 125–144.
- ZENG, J. (2017): "Forecasting aggregates with disaggregate variables: Does boosting help to select the Most relevant predictors?," *Journal of Forecasting*, 36(1), 74–90.
- ZOU, H. (2006): "The Adaptive Lasso and Its Oracle Properties," *Journal of the American Statistical Association*, 101(476), 1418–1429.